colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year

colour — First Iter — Last Iter

Japan LL Survey | NOAA LL AI Survey | NOAA LL BS Survey | NOAA LL GOA Survey

Survey Ages

Survey Lengths

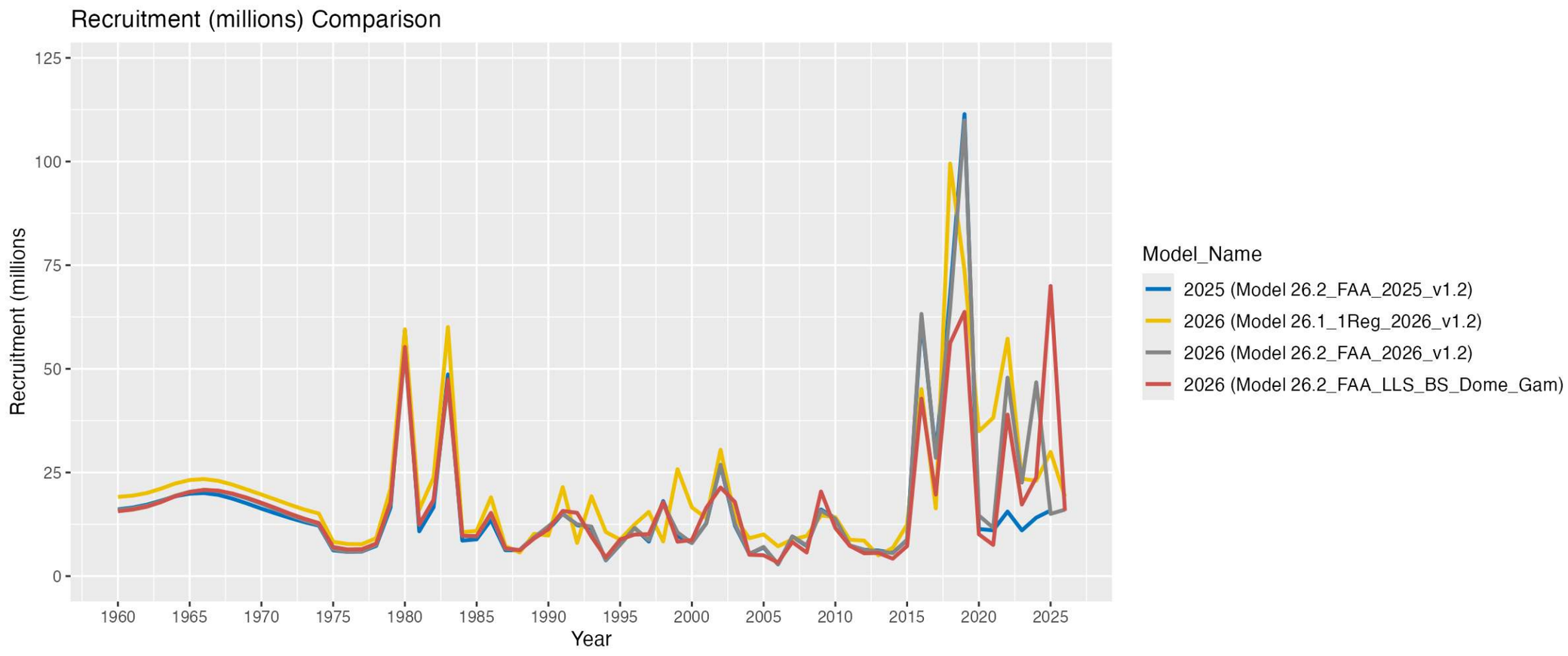
Mean Composition Bin

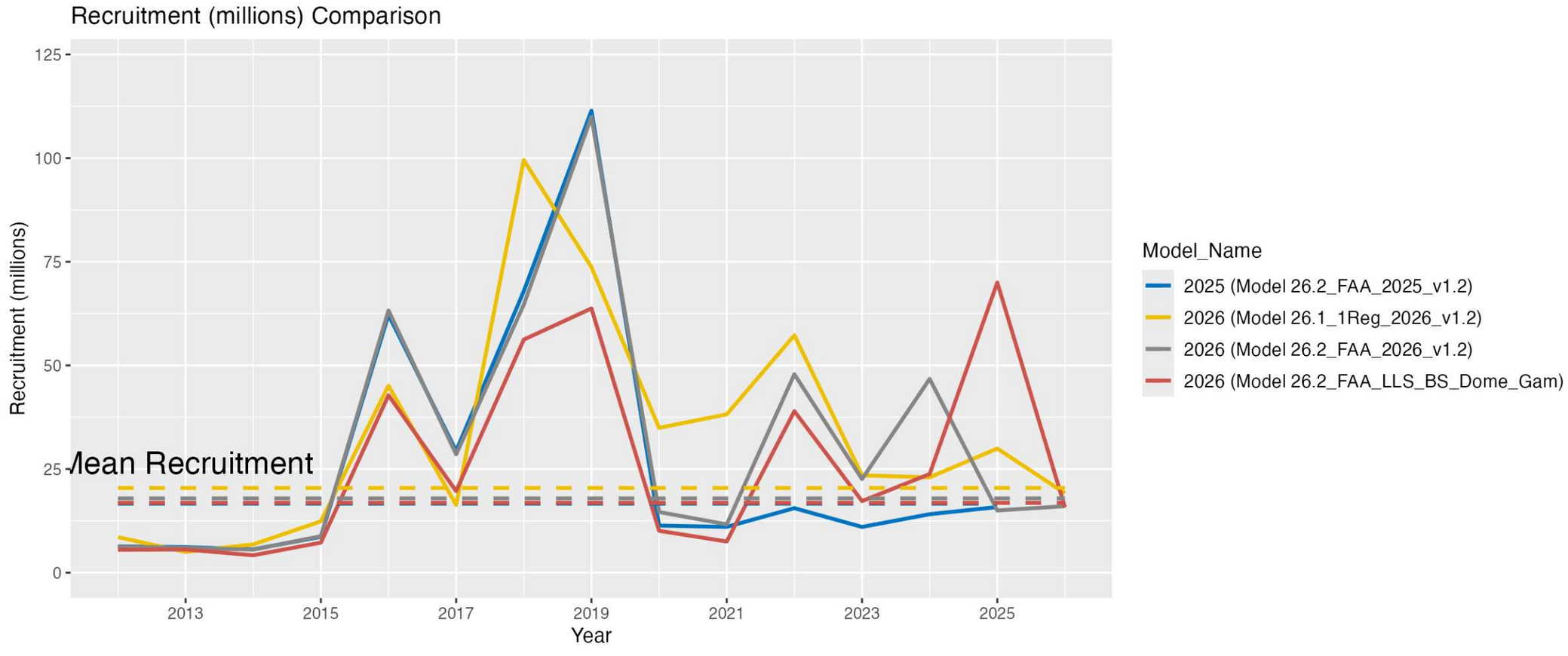
10.0
7.5
5.0

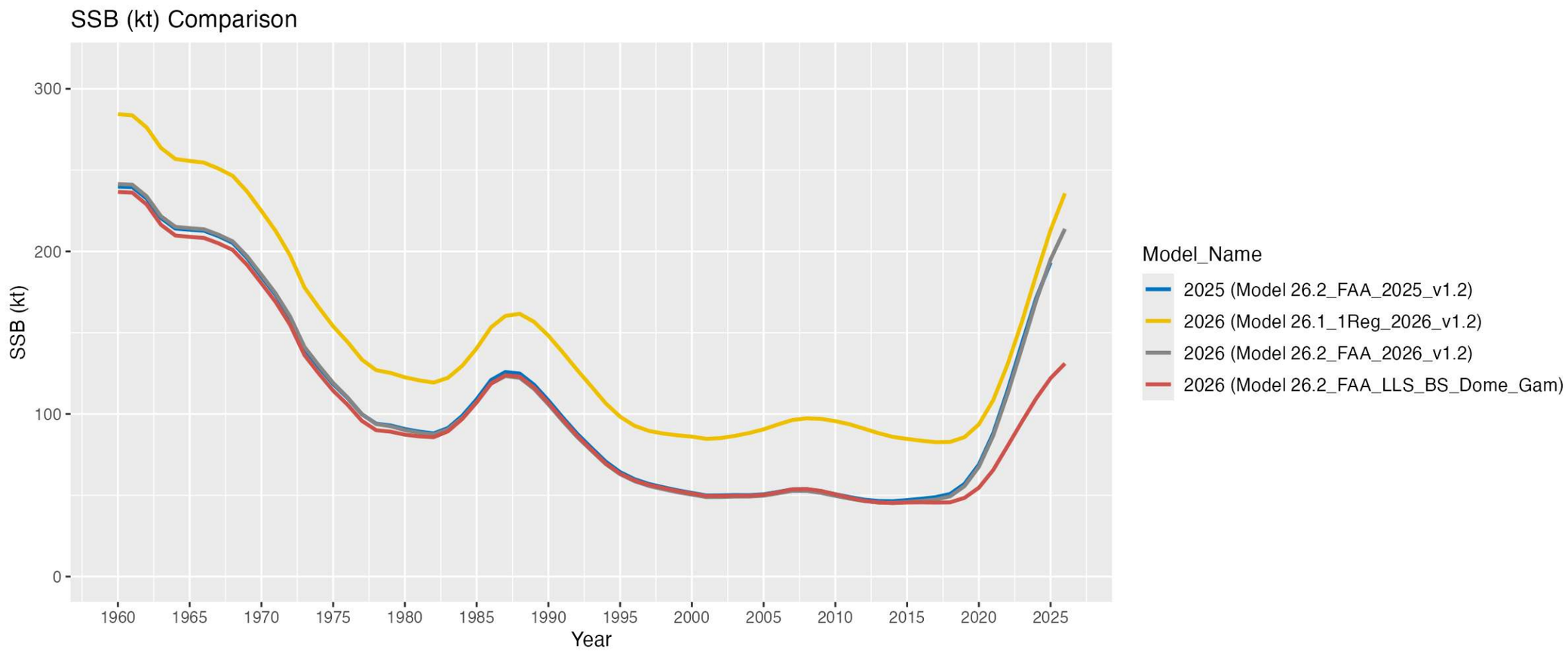
67.5
65.0
62.5

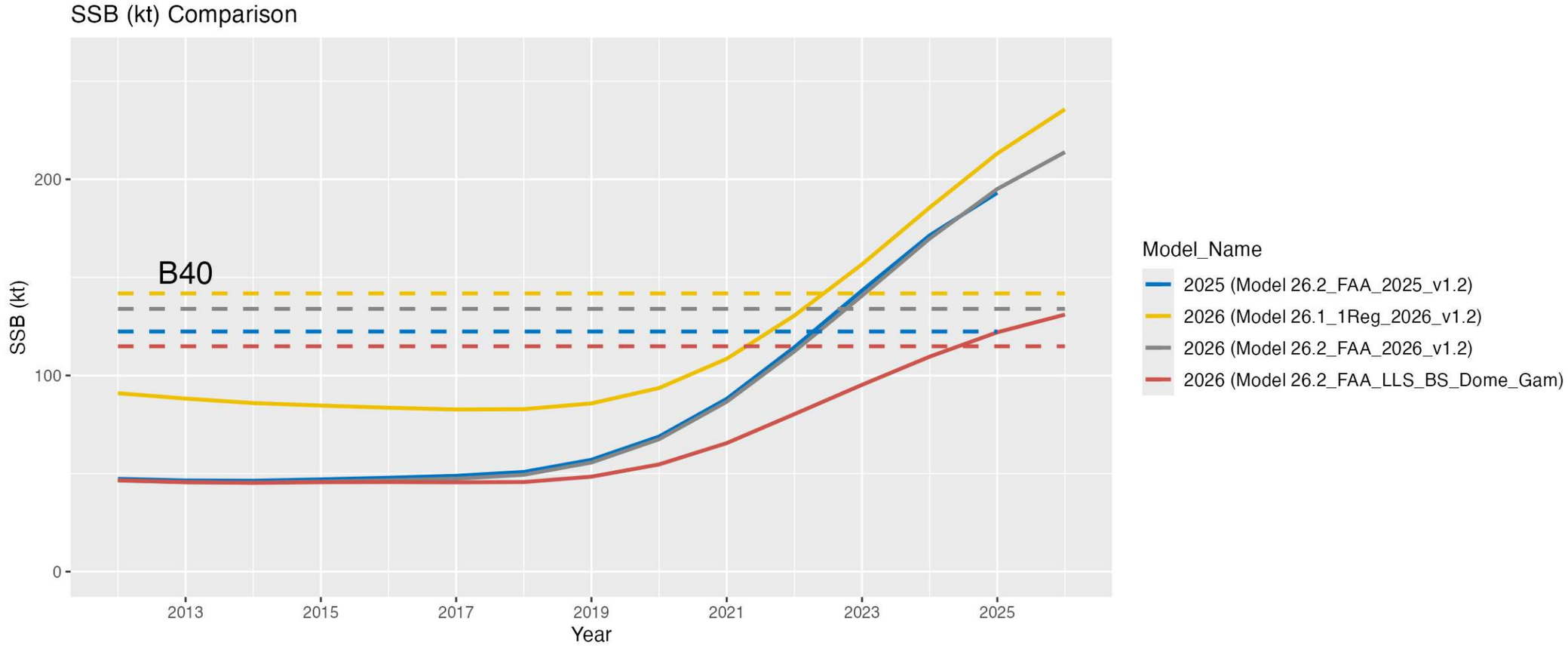
1980 1990 2000 2010 2020

Year


Recruitment (millions) Comparison
Recruitment (millions
125
100
75
50
25
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_LLS_BS_Dome_Gam)


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_LLS_BS_Dome_Gam)


SSB (kt) Comparison
300
200
100
0
SSB (kt)
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_LLS_BS_Dome_Gam)

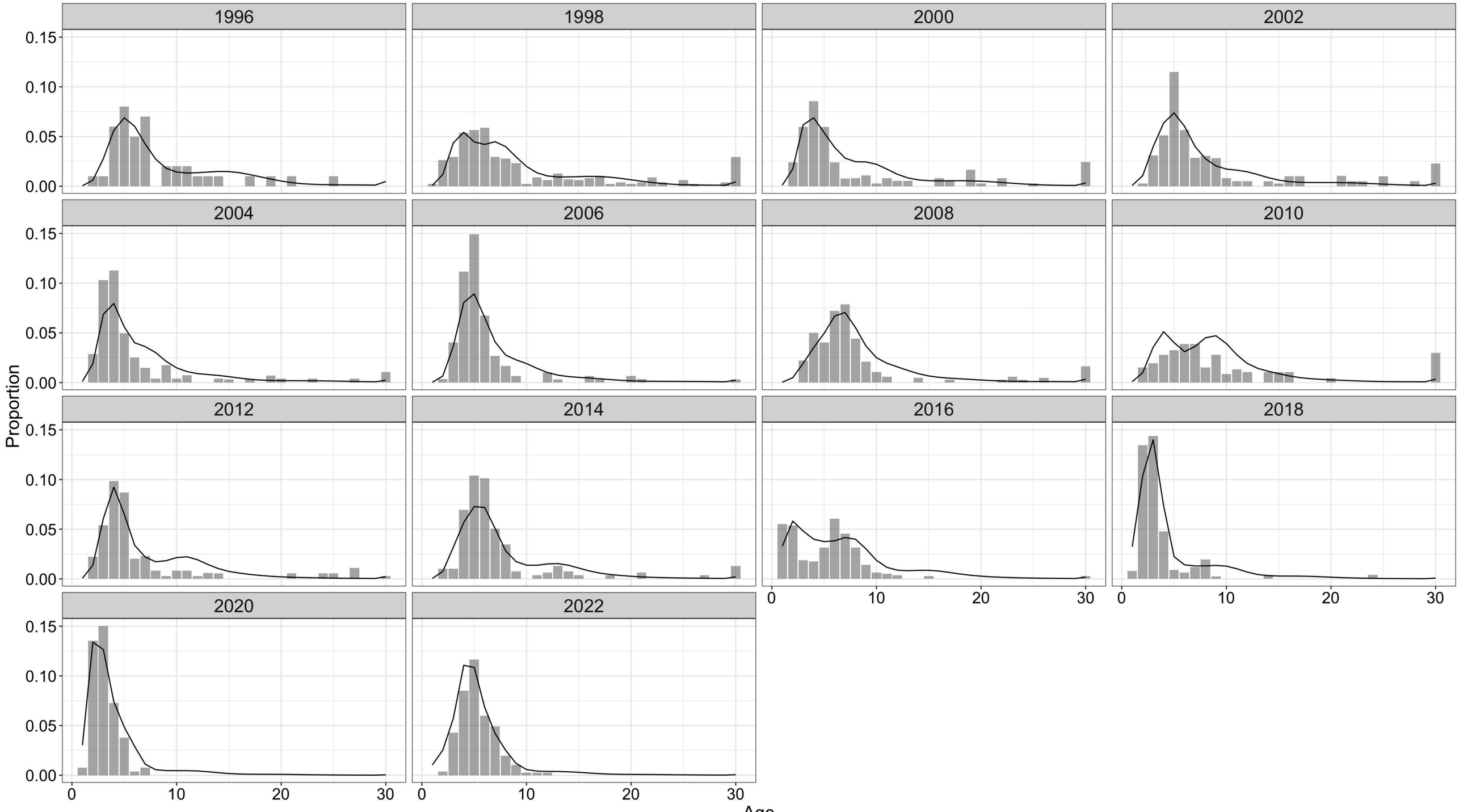

SSB (kt) Comparison
B40
200
100
0
SSB (kt)
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_LLS_BS_Dome_Gam)

AI LL Survey Age Comps Female AK

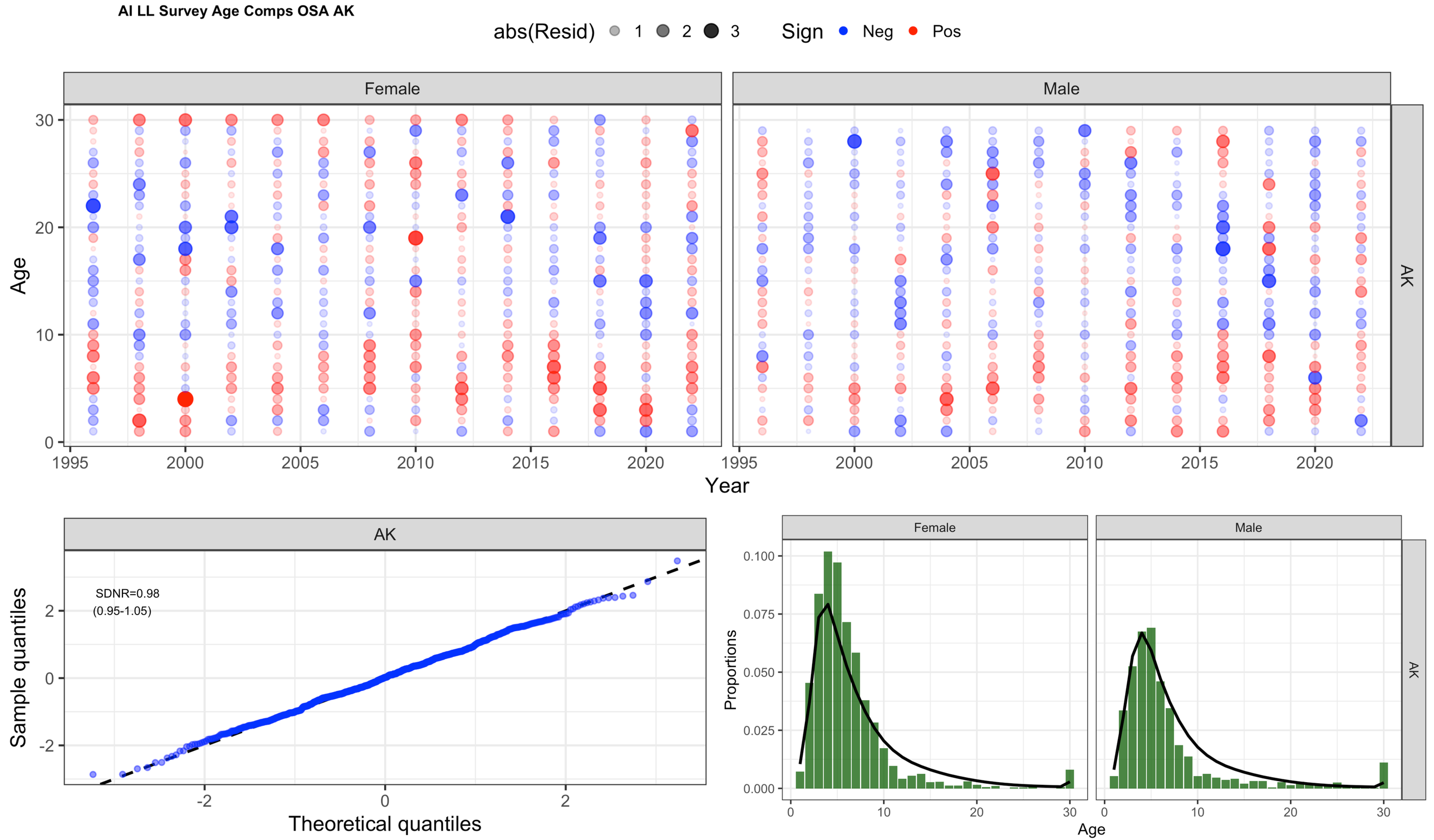
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022

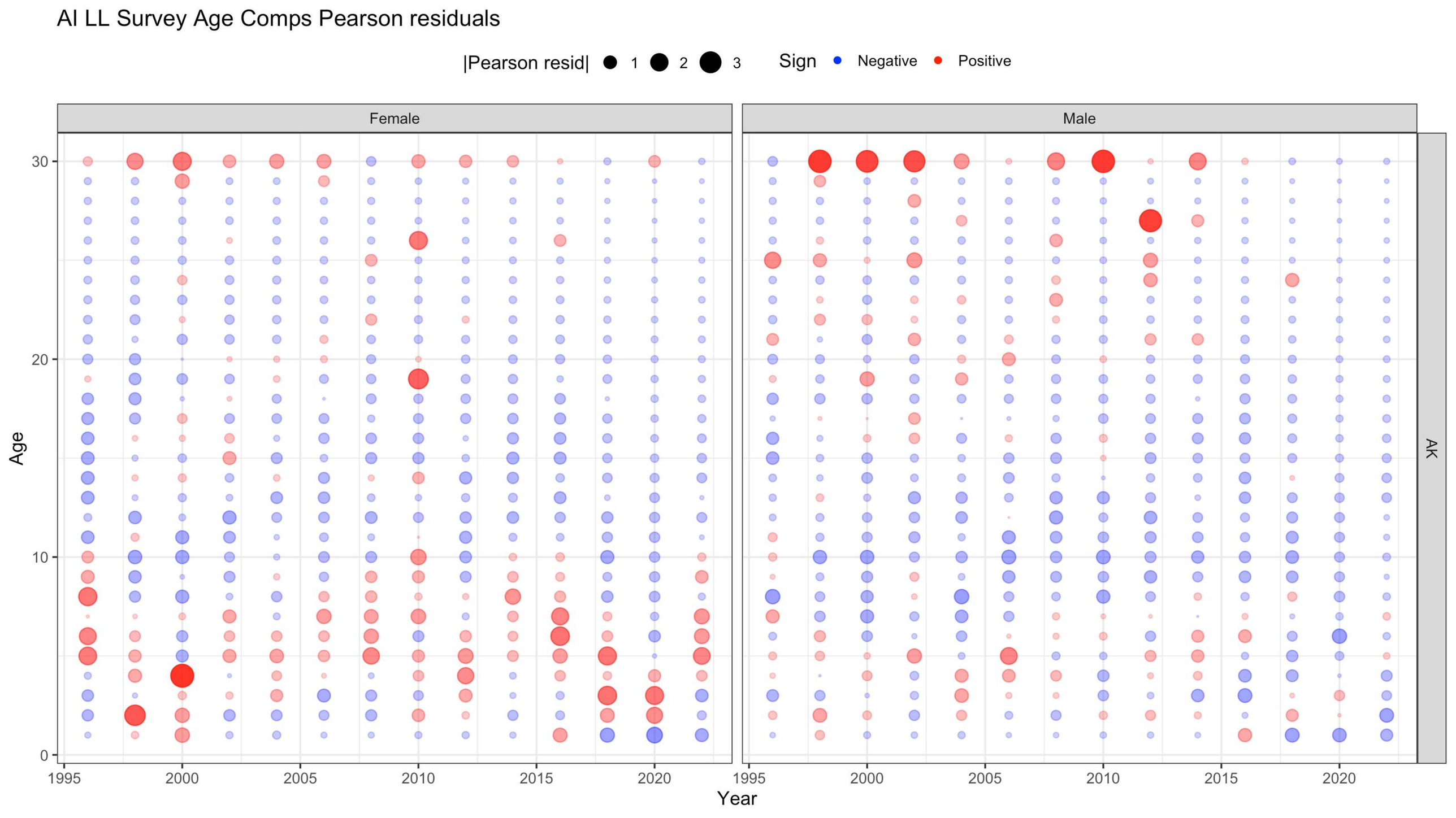
Proportion
0.00
0.05
0.10
0.15
0.20
0
10
20
30
Age

AI LL Survey Age Comps Male AK



AI LL Survey Age Comps OSA AK




AI LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020

Model 26.2_FAA_LLS_BS_Dome_Gam Population Population 1

Female | Male | AK

Spawning Weight-at-Age (kg)

0.075
0.050
0.025
0.000

0 10 20 30

Age

Model 26.2_FAA_LLS_BS_Dome_Gam Population Population 1

Female

Male

AK

Maturity at Age

1.00

0.75

0.50

0.25

0.00

0

10

20

30

Age

Model 26.2_FAA_LLS_BS_Dome_Gam Population Population 1

Female

Male

AK

Spawning Weight at Age

6

4

2

0

0

10

20

30

Age

BS LL Survey Age Comps Female AK

1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion
0.0 0.1 0.2 0.3
0 10 20 30
Age

BS LL Survey Age Comps Male AK

1997 1999 2001 2003

2005 2007 2009 2011

2013 2015 2017 2019

2021 2023

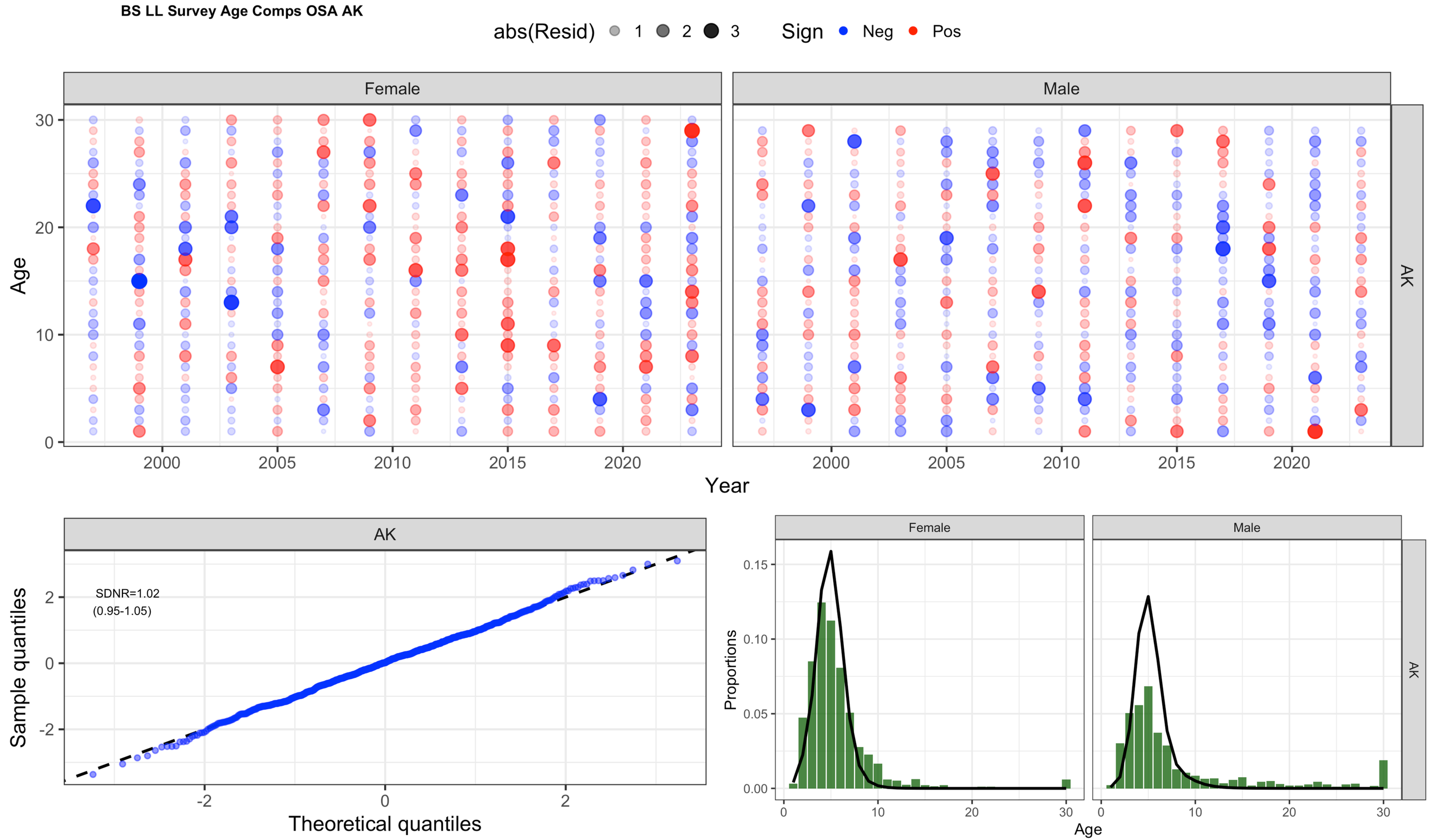
Proportion

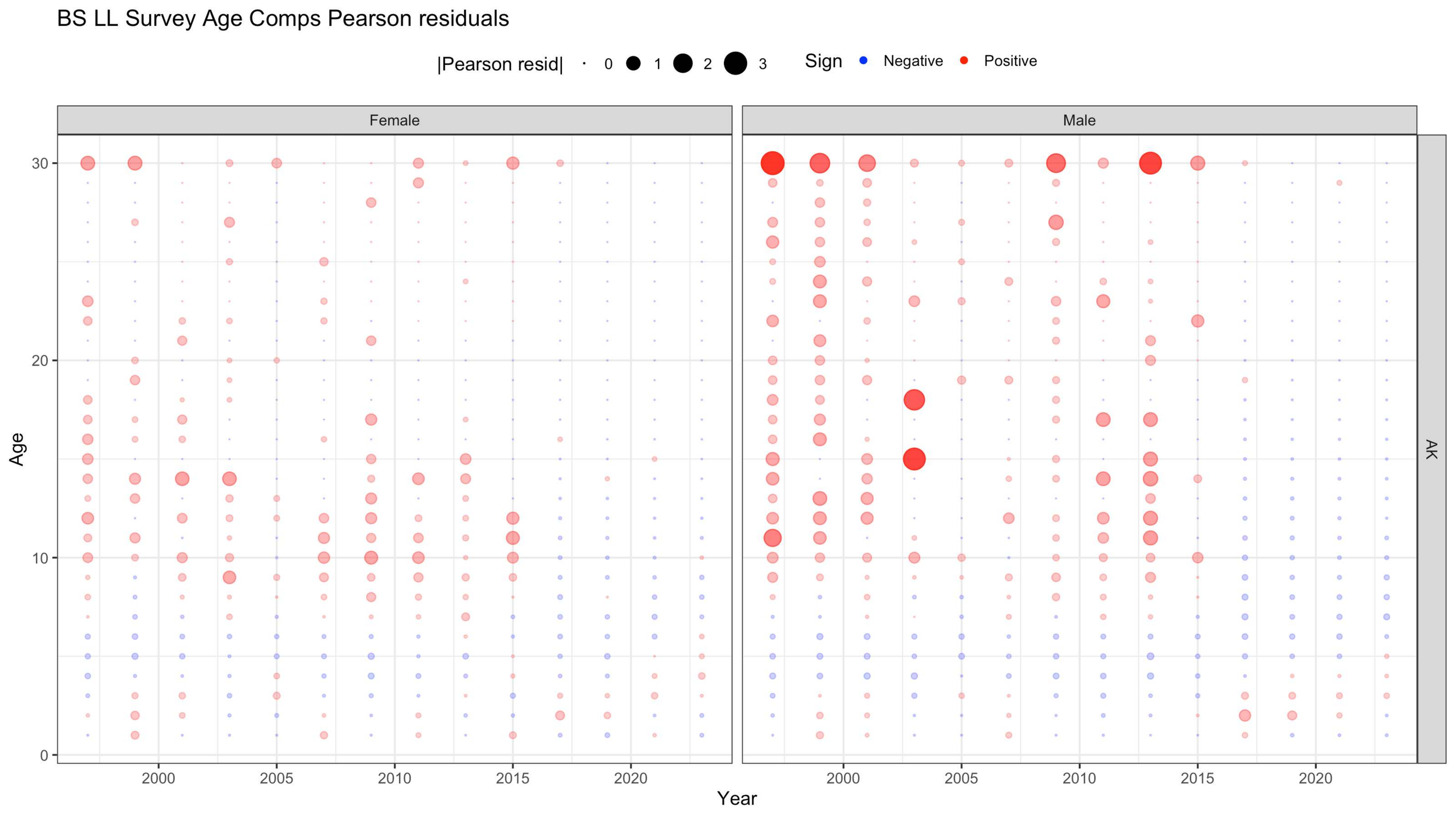
0.00 0.05 0.10 0.15

0 10 20 30

Age

BS LL Survey Age Comps OSA AK




BS LL Survey Age Comps Pearson residuals
|Pearson resid|
0
1
2
3
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020

Model 26.2_FAA_LLS_BS_Dome_Gam
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

AK

26.2_FAA_LLS_BS_Dome_Gam

JPN LLS Lengths

JPN LLS Index

GOA LLS Index

AI LLS Index

BS LLS Index

JPN LLS Ages

GOA LLS Ages

AI LLS Ages

BS LLS Ages

TF Lengths

TF Catch

LLF Catch

LLF Ages

1960

1980

2000

2020

Year

GOA LL Survey Age Comps Female AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

Proportion

0.0 0.1 0.2

0 10 20 30

Age

GOA LL Survey Age Comps Male AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

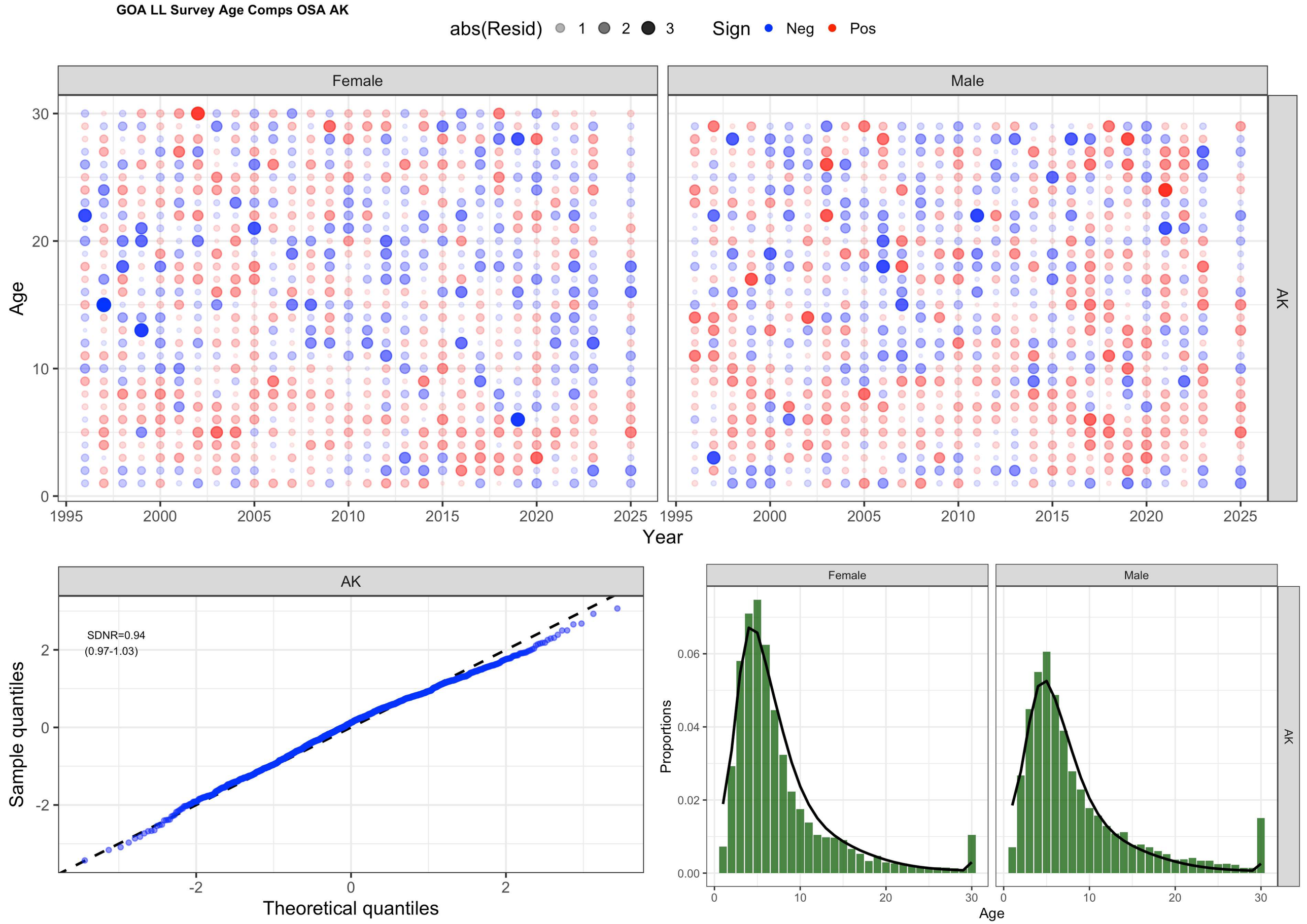
Proportion

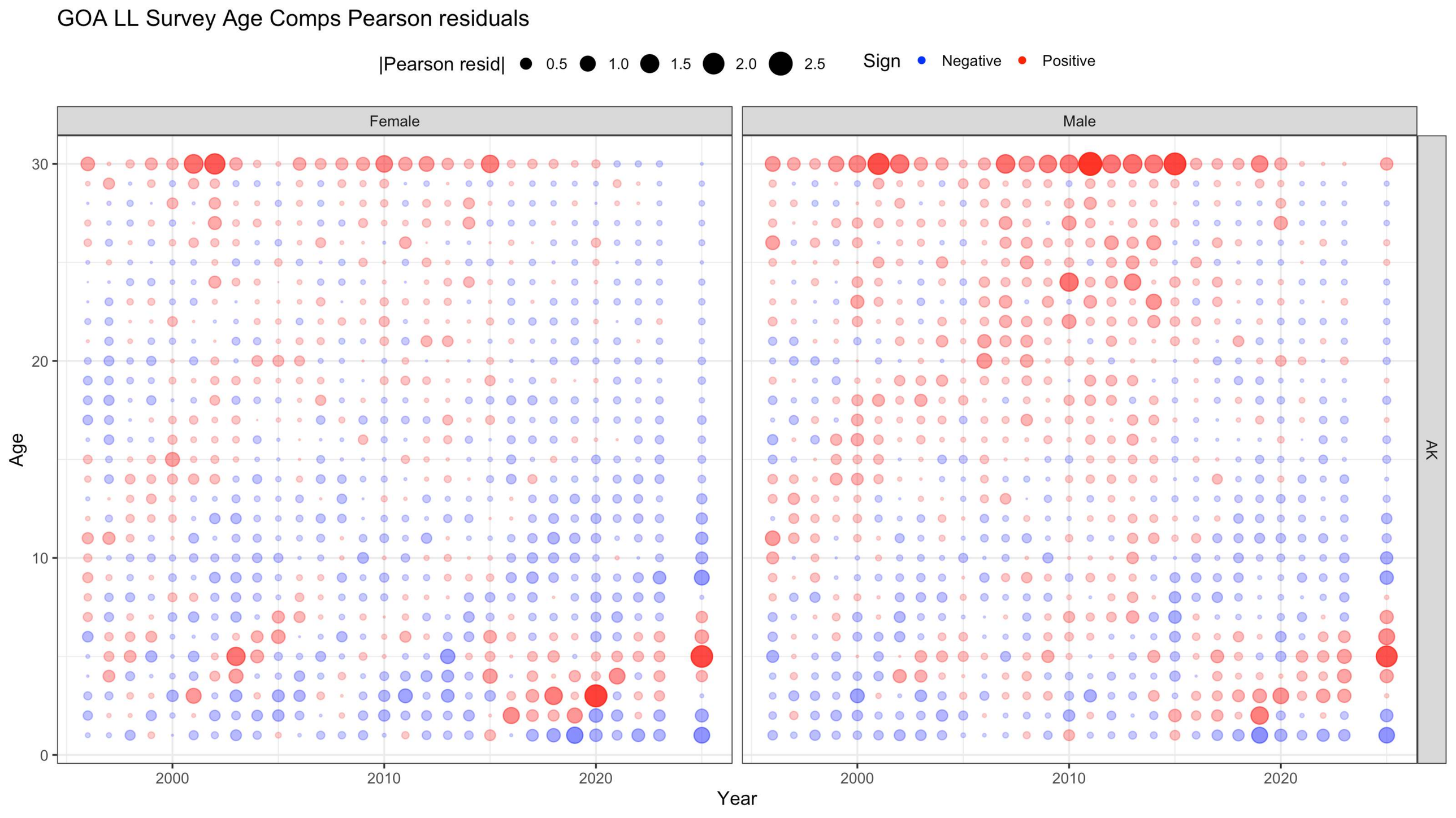
0.00 0.05 0.10 0.15

0 10 20 30

Age

GOA LL Survey Age Comps OSA AK




GOA LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
2.5
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

Model — 26.2_FAA_LLS_BS_Dome_Gam

BS LLS

AI LLS

GOA LLS

JPN LLS

RPNs (1000s)

Year

Jitter SSB and Recruitment

Region 1

Recruitment (millions)

SSB (kt)

Proportion Converged: 1
Proportion jnLL < original jnLL: 0.012

Hessian

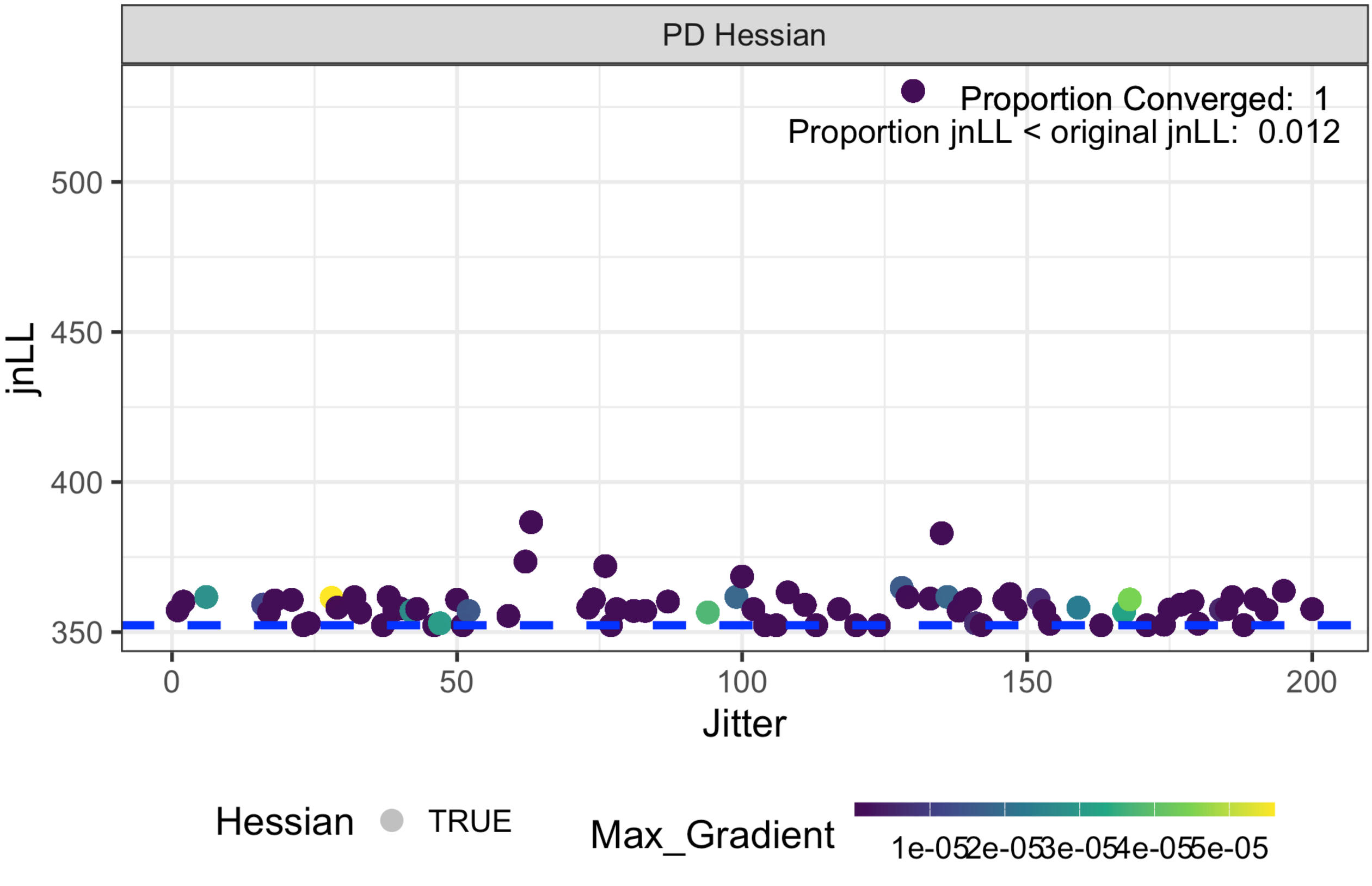
TRUE

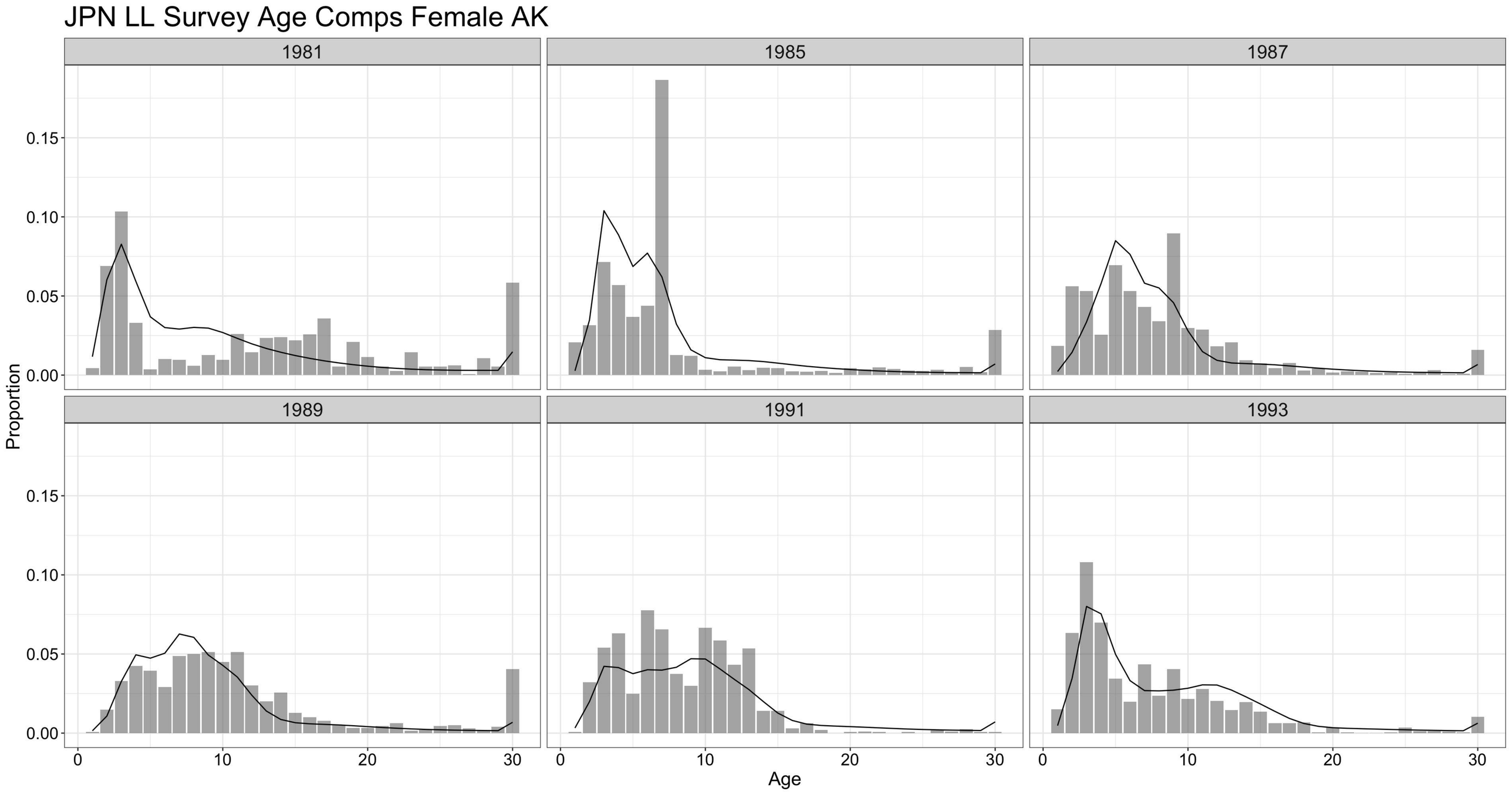
Value

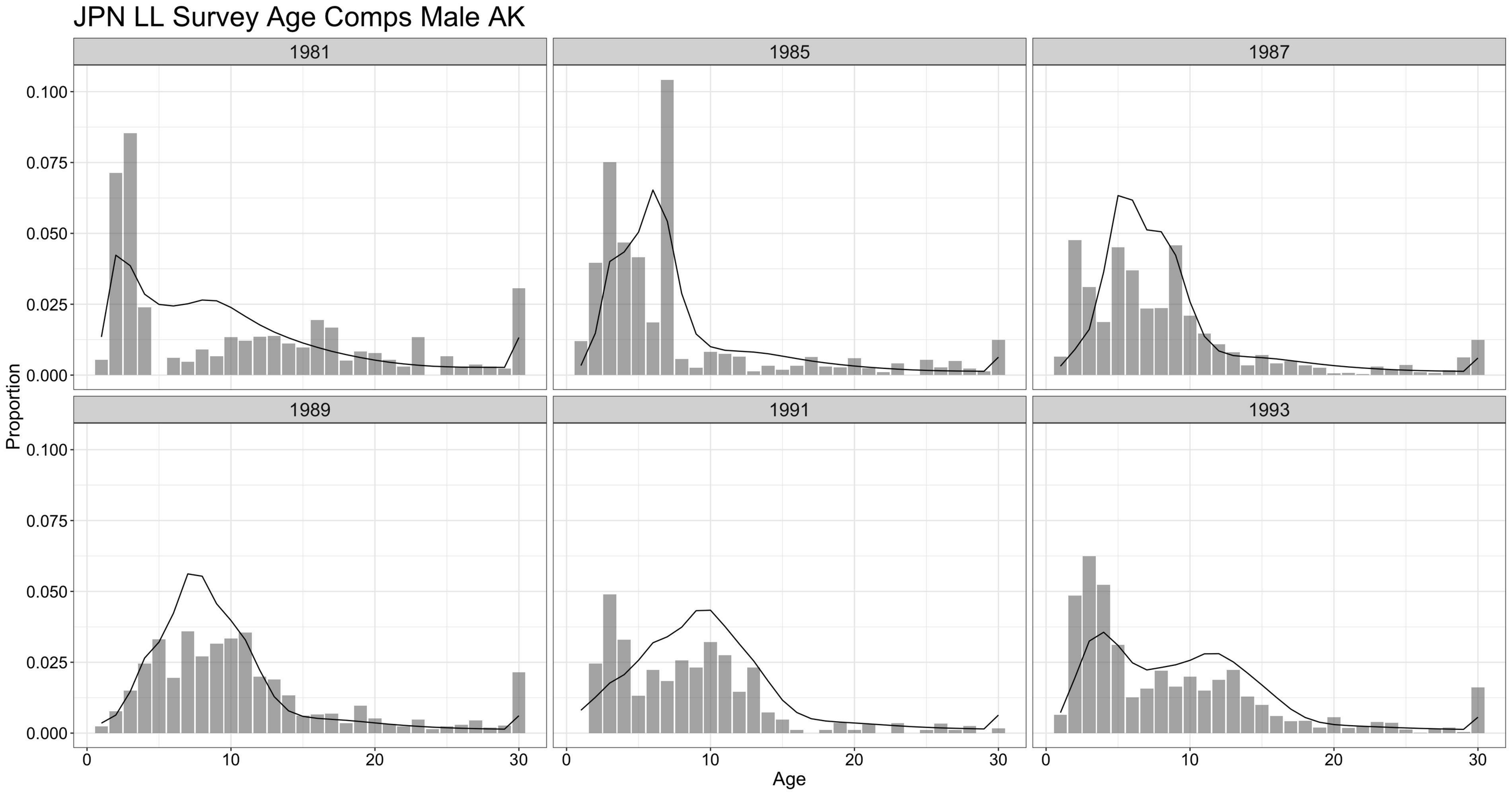
Year

100
50
0
300
200
100
0
1960
1980
2000
2020

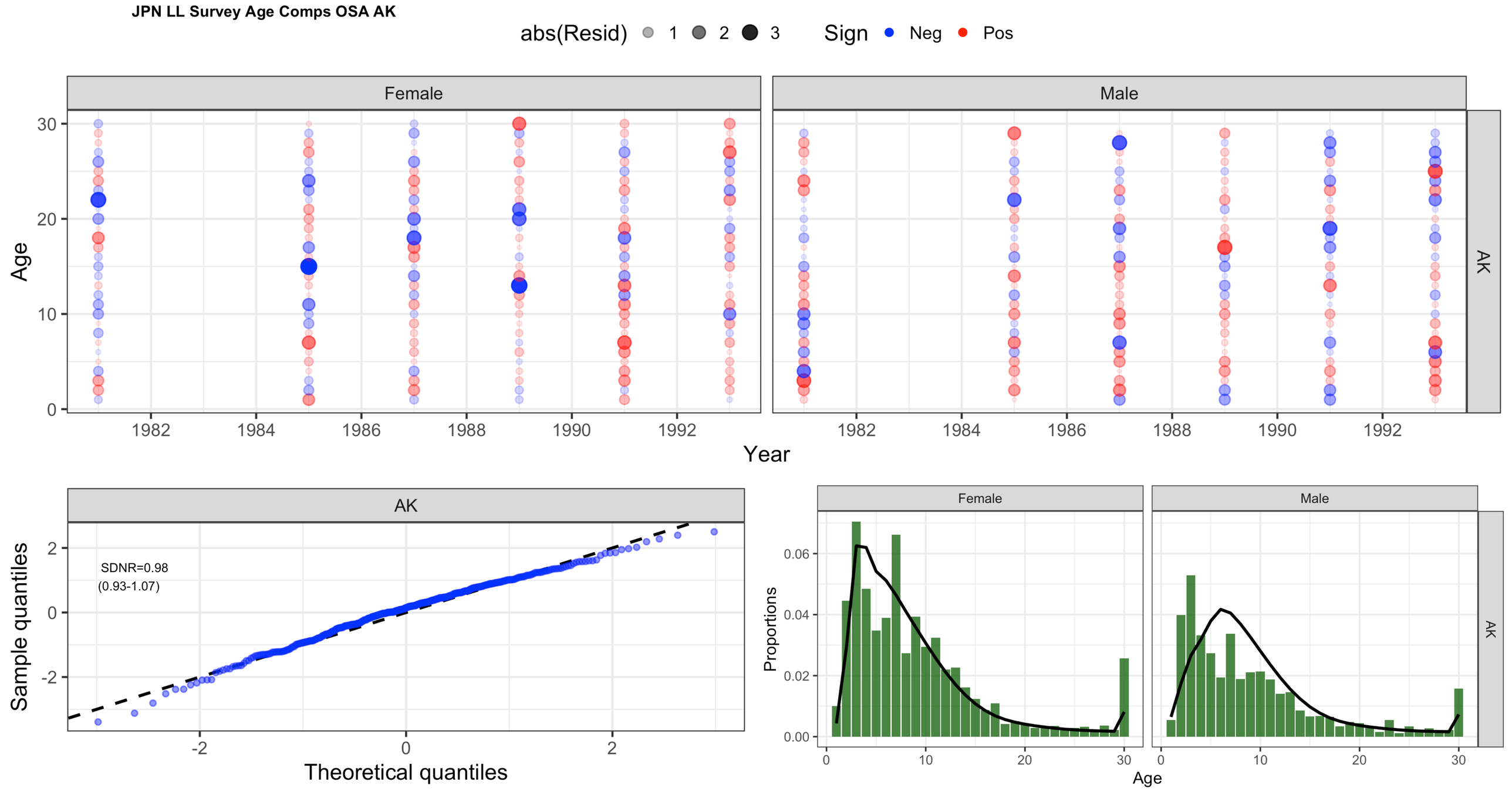
Jitter nLL and Convergence

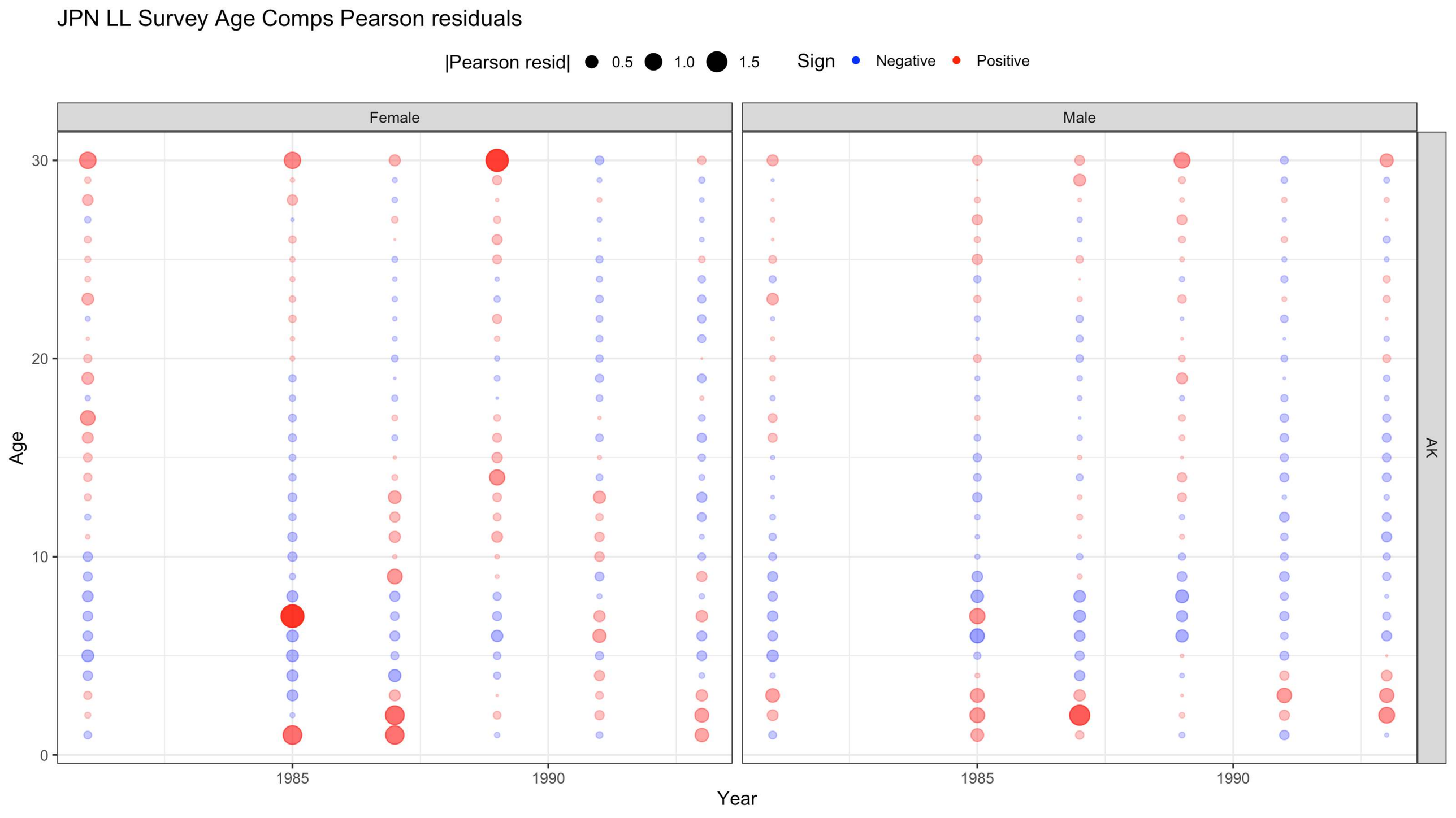



JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
0.100
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age

JPN LL Survey Age Comps OSA AK




JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year

JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

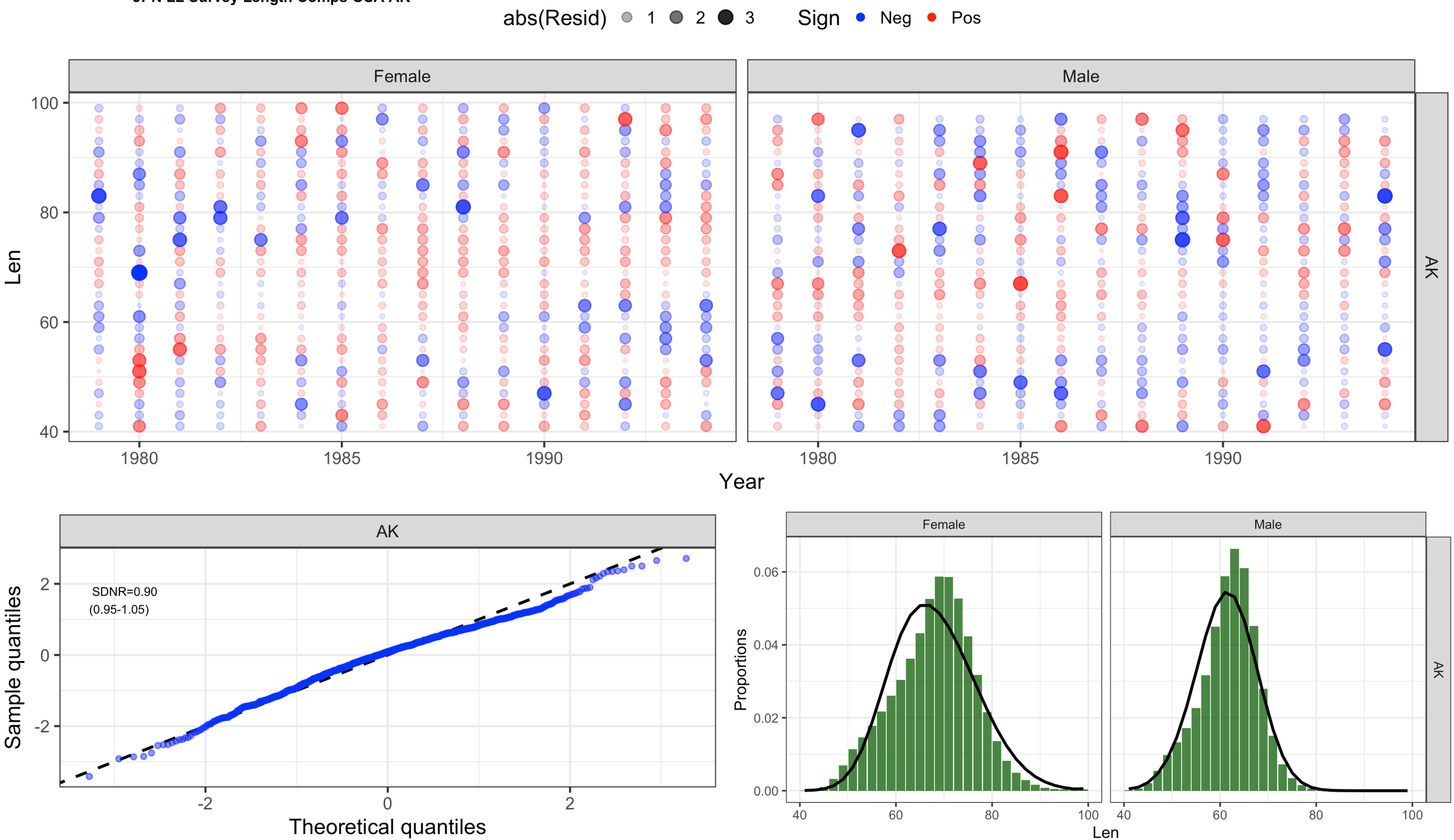
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

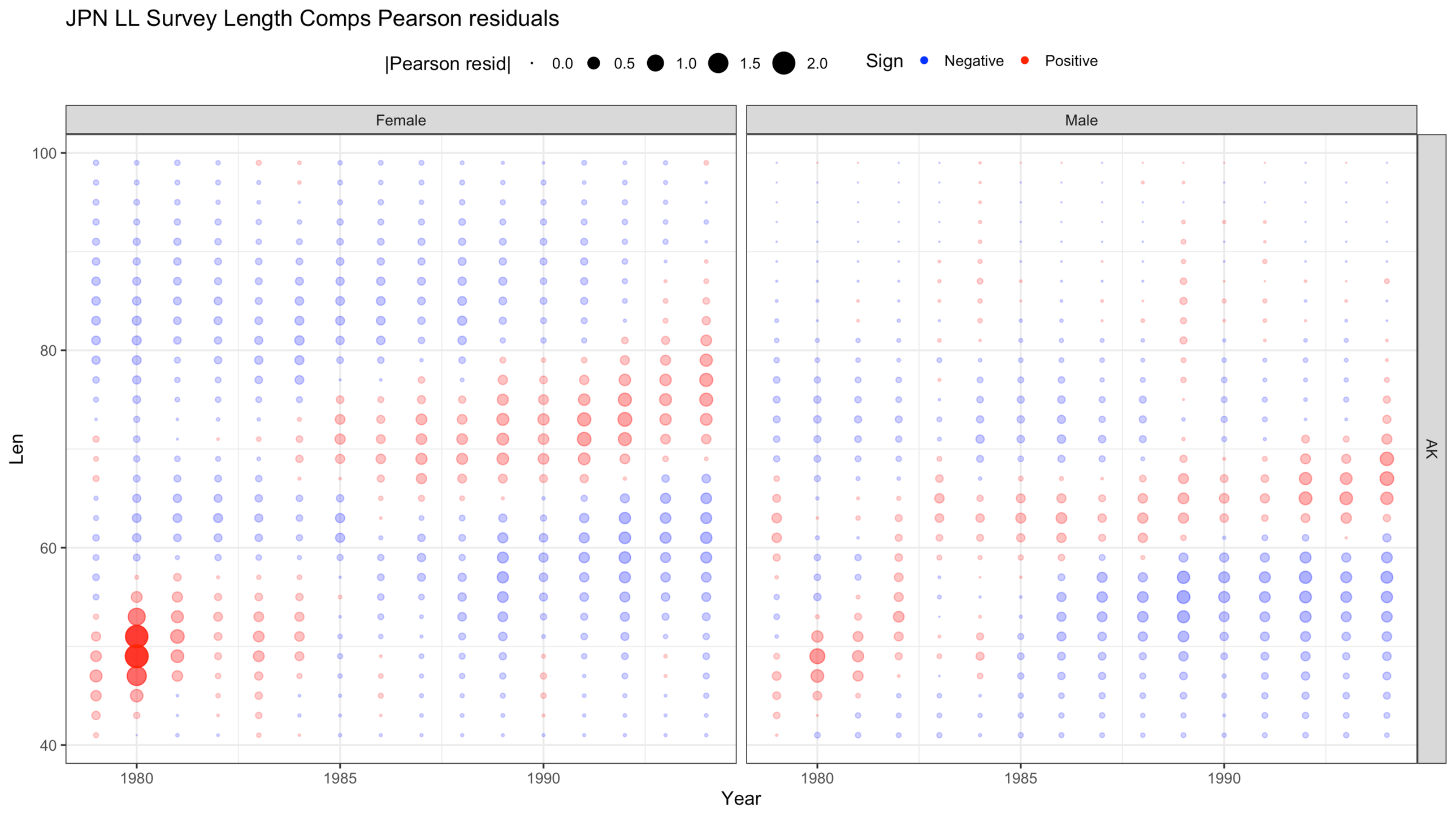
JPN LL Survey Length Comps Male AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

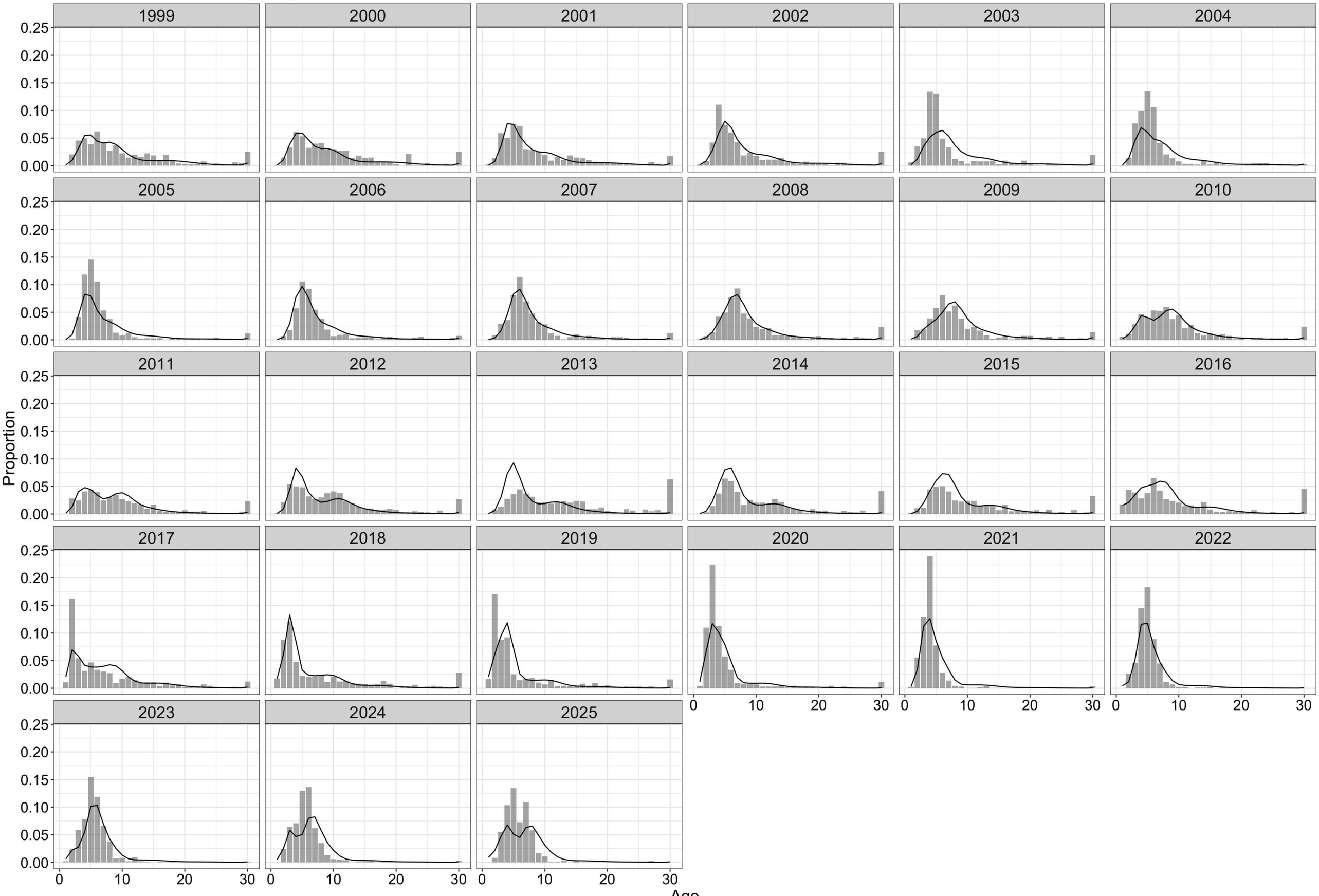
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps OSA AK

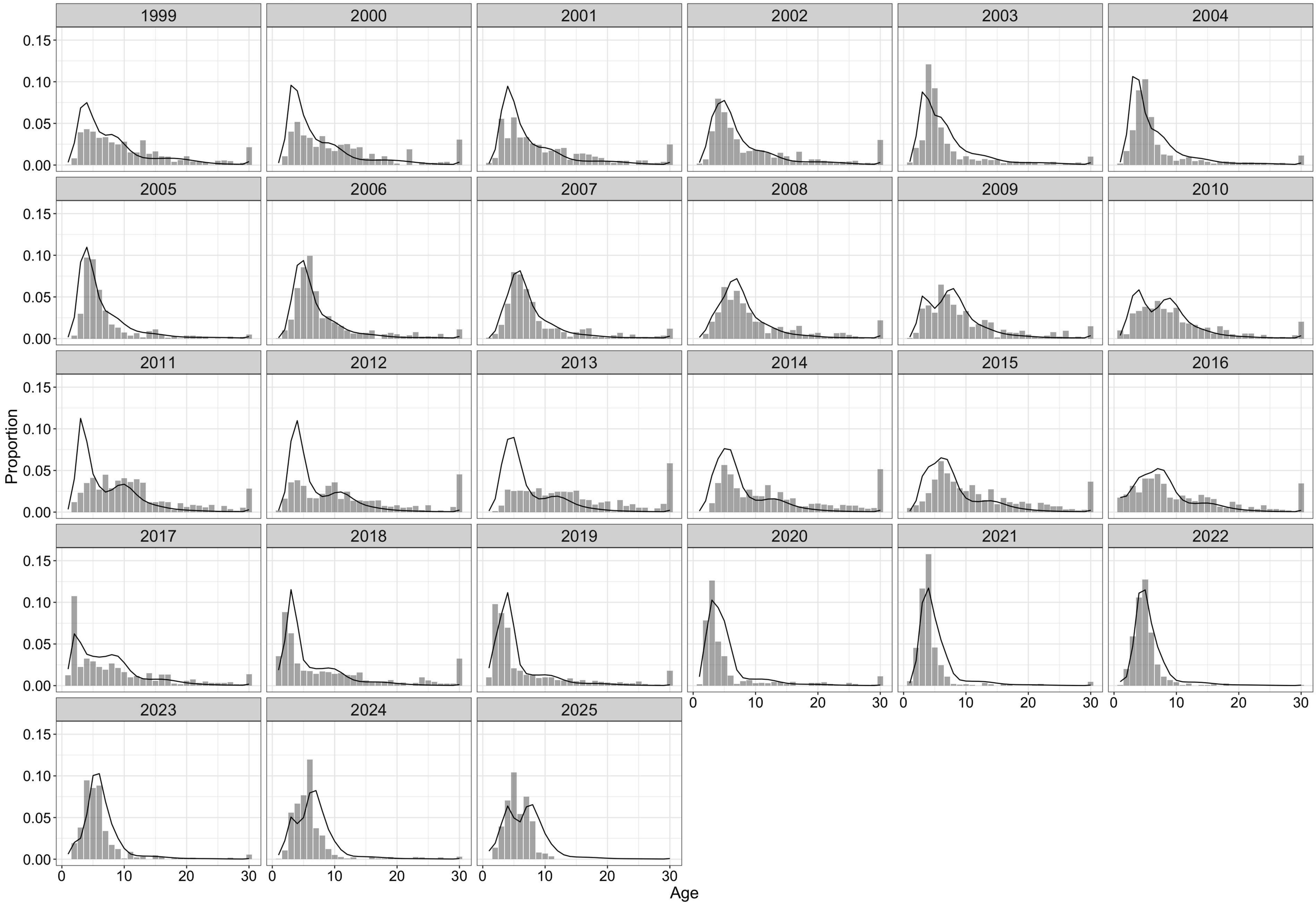



JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

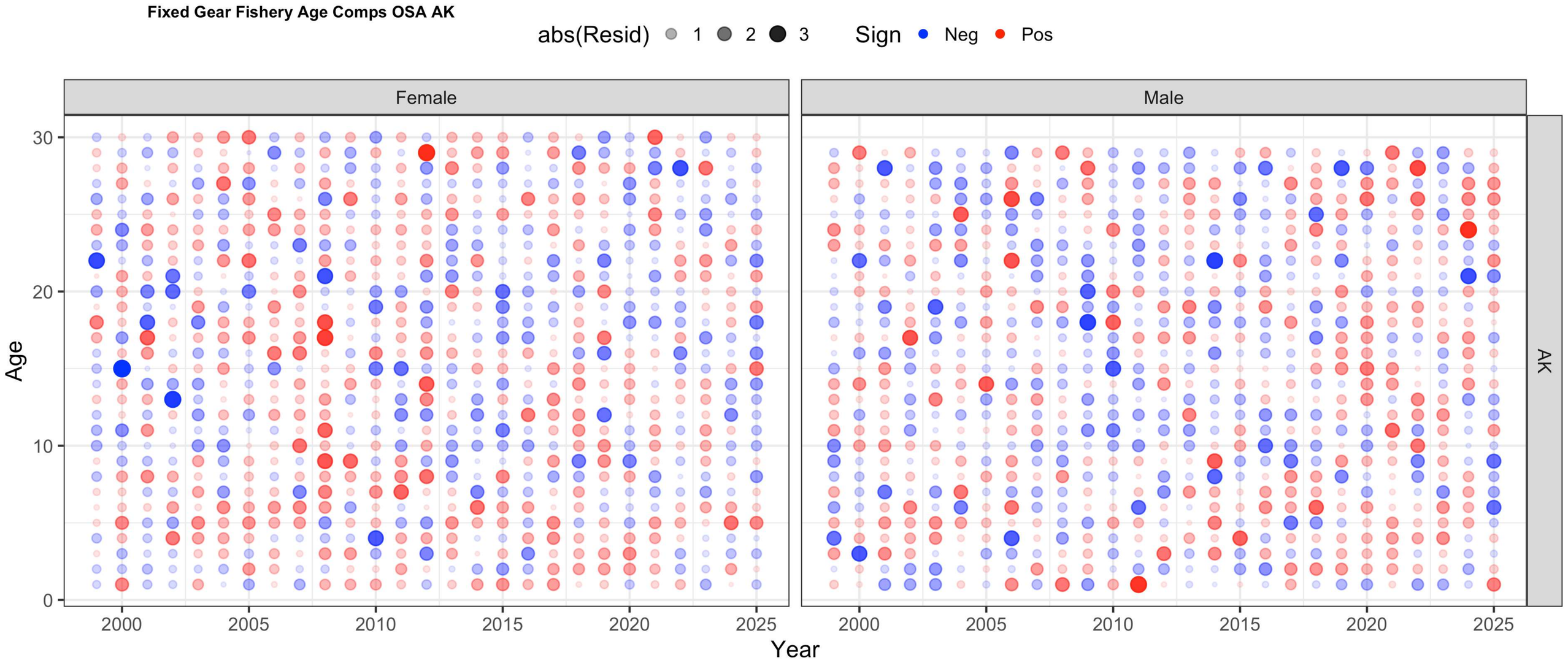
Fixed Gear Fishery Age Comps Female AK

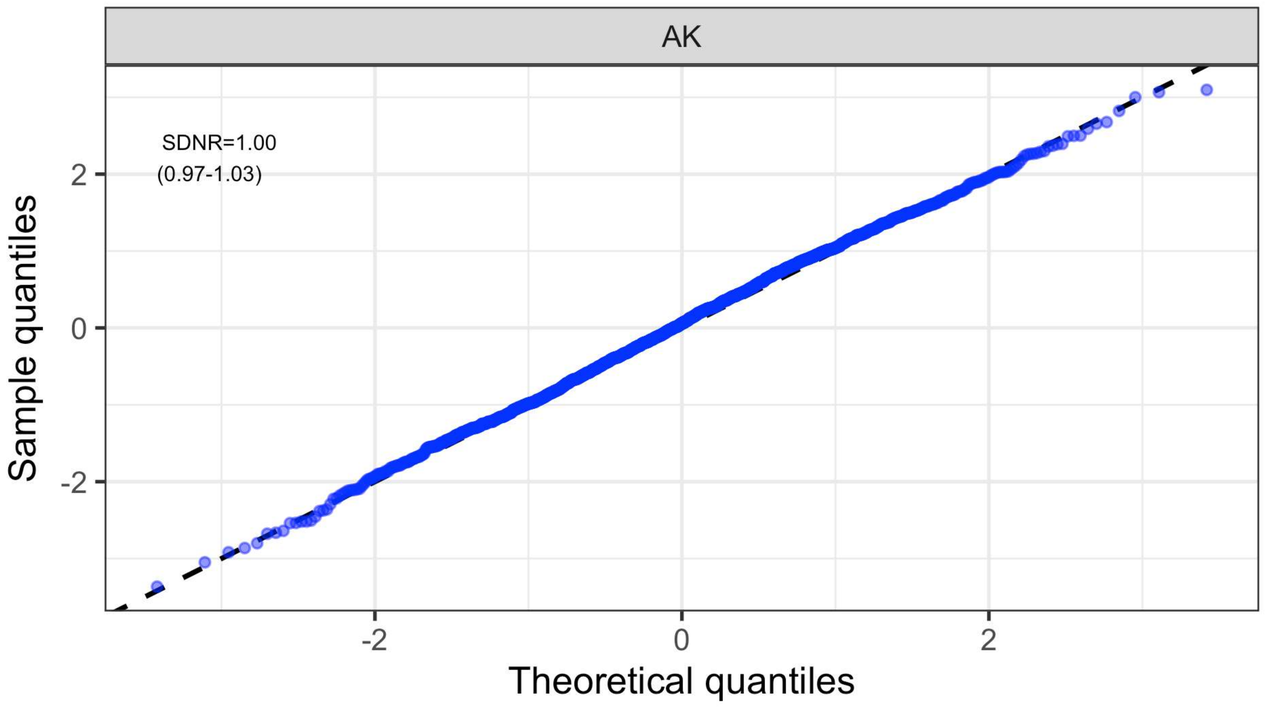


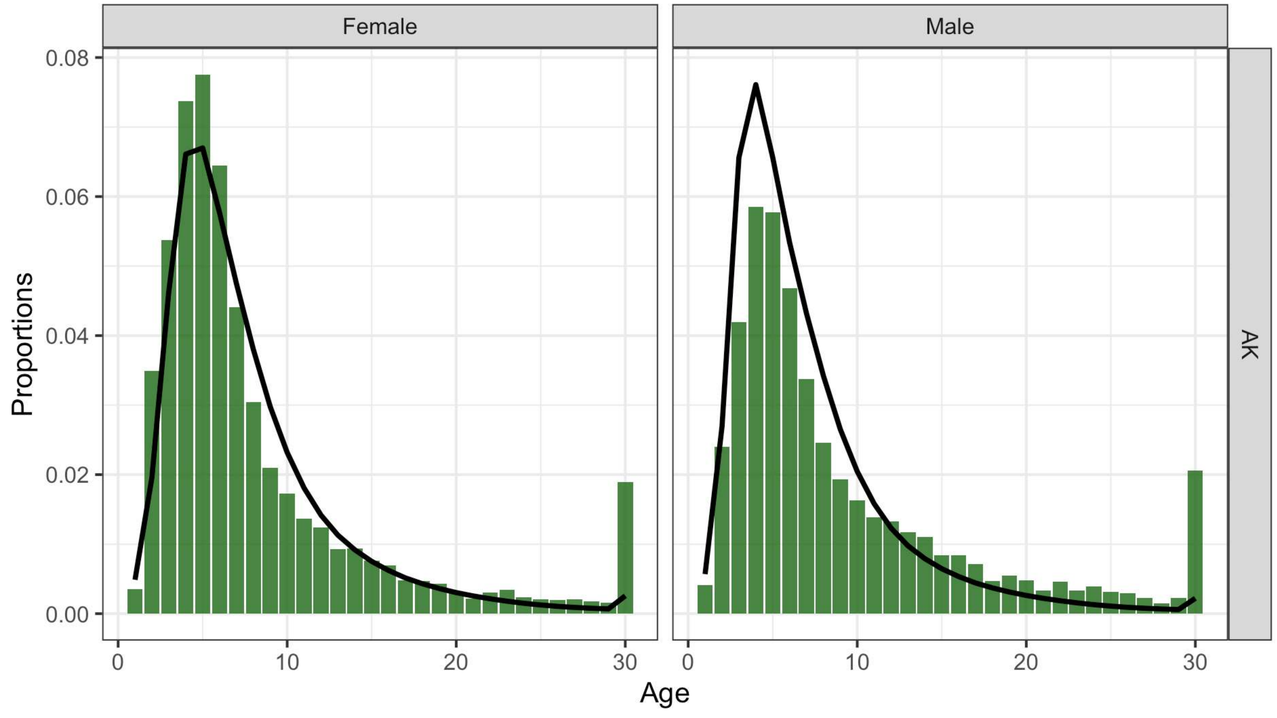
Fixed Gear Fishery Age Comps Male AK



Fixed Gear Fishery Age Comps OSA AK







Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
1
2
3
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

Type: Age, Catch, Index, jnLL, Length, Penalty, Prior

26.2_FAA_LLS_BS_Dome_Gam

Likelihood

200

0

-200

Catch nLL

Fishery Age nLL

Fishery Length nLL

Fishing Mortality Penalty

Initial Age Penalty

jnLL

M Prior

Recruitment Penalty

Selectivity Penalty

Survey Age nLL

Survey Index nLL

Survey Length nLL

Likelihood Component

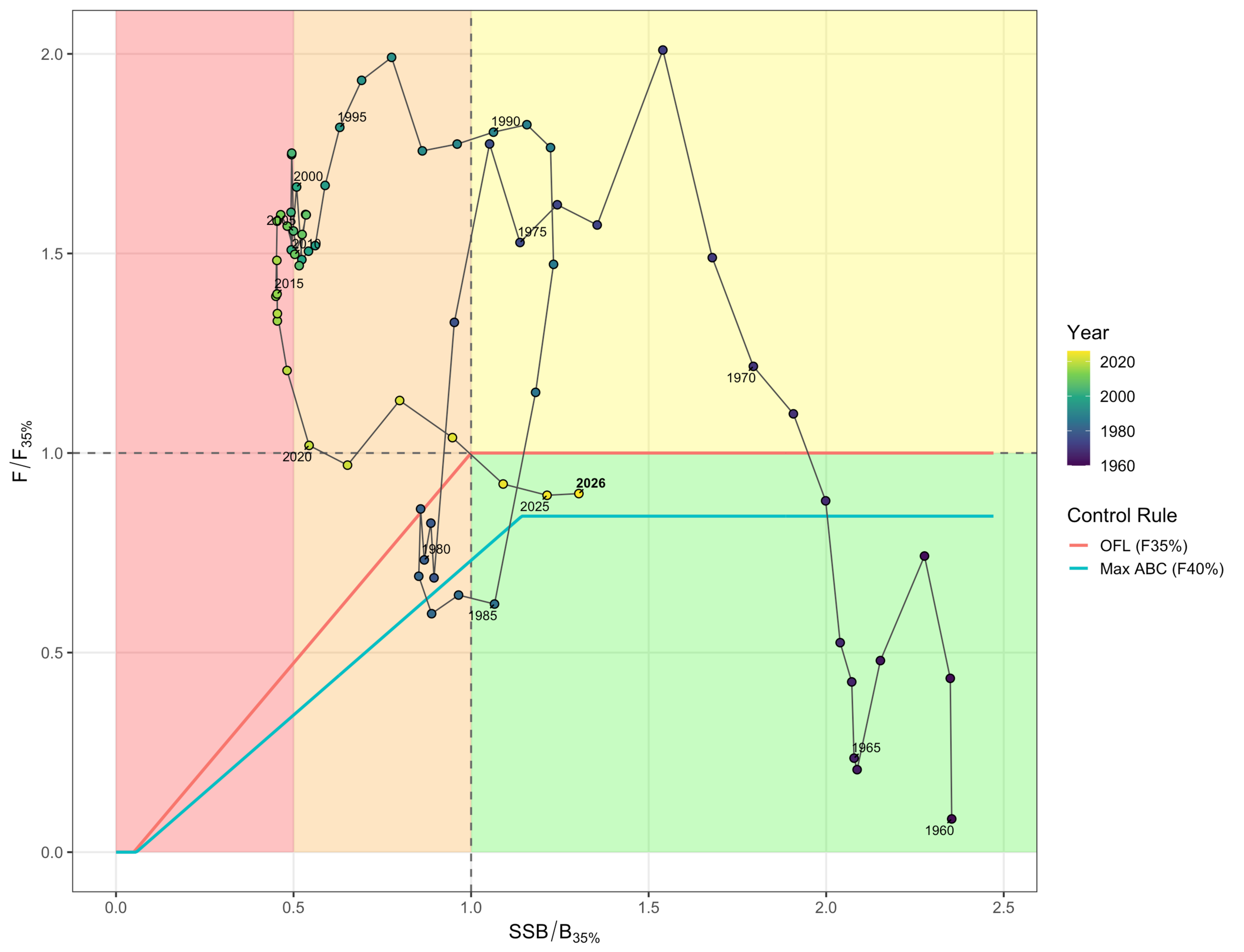
Numbers-at-Age

Sex — Female ···· Male

Age- 1
Age- 2
Age- 3
Age- 4
Age- 5
Age- 6
Age- 7
Age- 8
Age- 9
Age- 10
Age- 11
Age- 12
Age- 13
Age- 14
Age- 15
Age- 16
Age- 17
Age- 18
Age- 19
Age- 20
Age- 21
Age- 22
Age- 23
Age- 24
Age- 25
Age- 26
Age- 27
Age- 28
Age- 29
Age- 30

Numbers-at-Age (millions)

Year


2.0
1.5
1.0
0.5
0.0
0.0
0.5
1.0
1.5
2.0
2.5
F/F35%
SSB/B35%
1995
1990
2000
2005
2010
1975
2015
1970
2020
2026
2025
1980
1985
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

Mohns Rho: -0.154

Mohns Rho: 0.1994

AK

Relative Difference from Terminal Year

1.0
0.5
0.0
-0.5

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

Year

0
20
40
60
80

0
50
100
150
200
250

1960
1980
2000
2020

AK

Recruitment (millions)

80
60
40
20

0
1
2
3
4
5
6
7

Years since cohort was last estimated

Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

Mohns Rho: -0.1535

Mohns Rho: 0.2284

AK

Relative Difference from Terminal Year

2
1
0

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

Year

0
20
40
60
80

0
50
100
150
200
250

1960
1980
2000
2020

AK

Recruitment (millions)

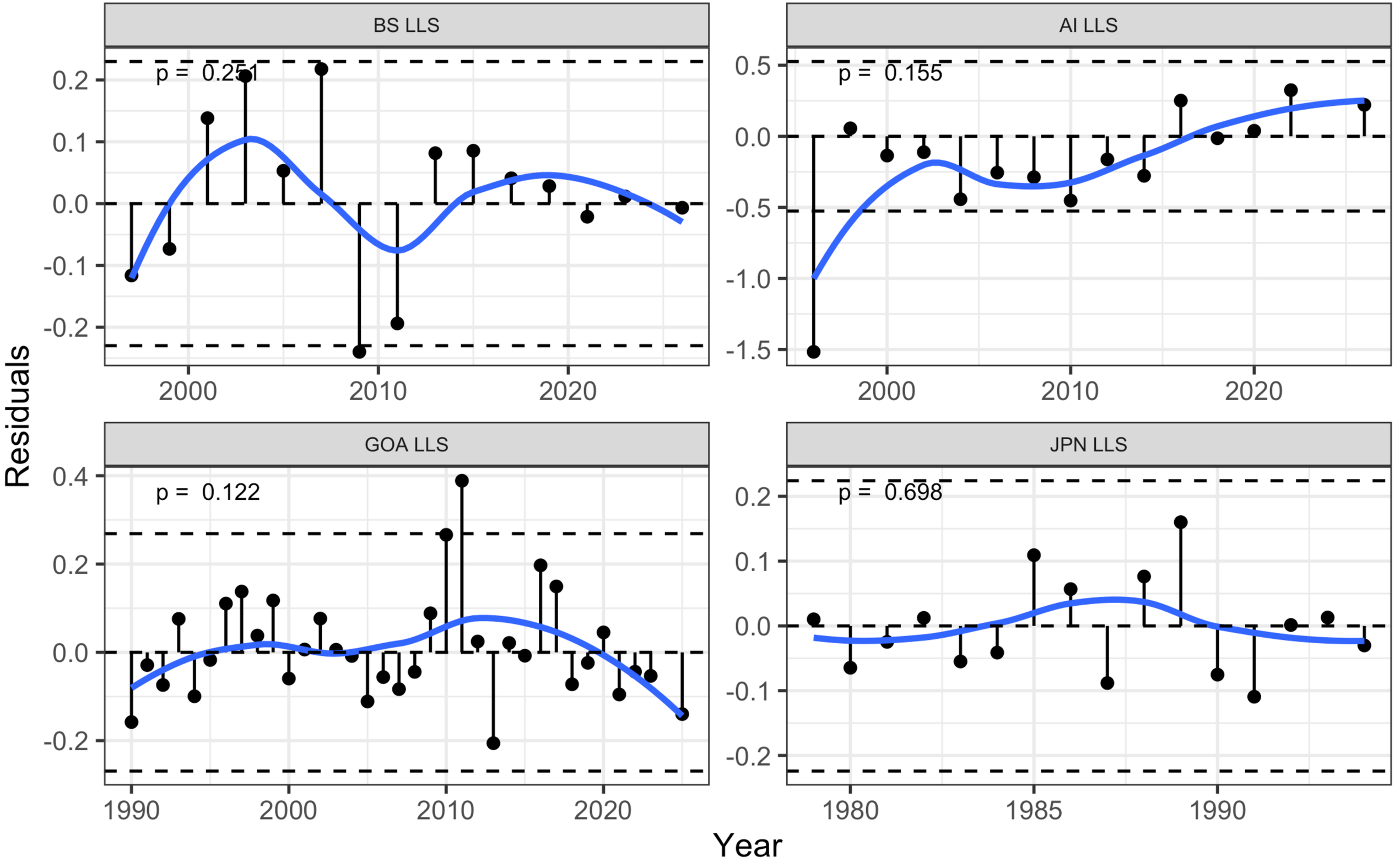
80
60
40
20

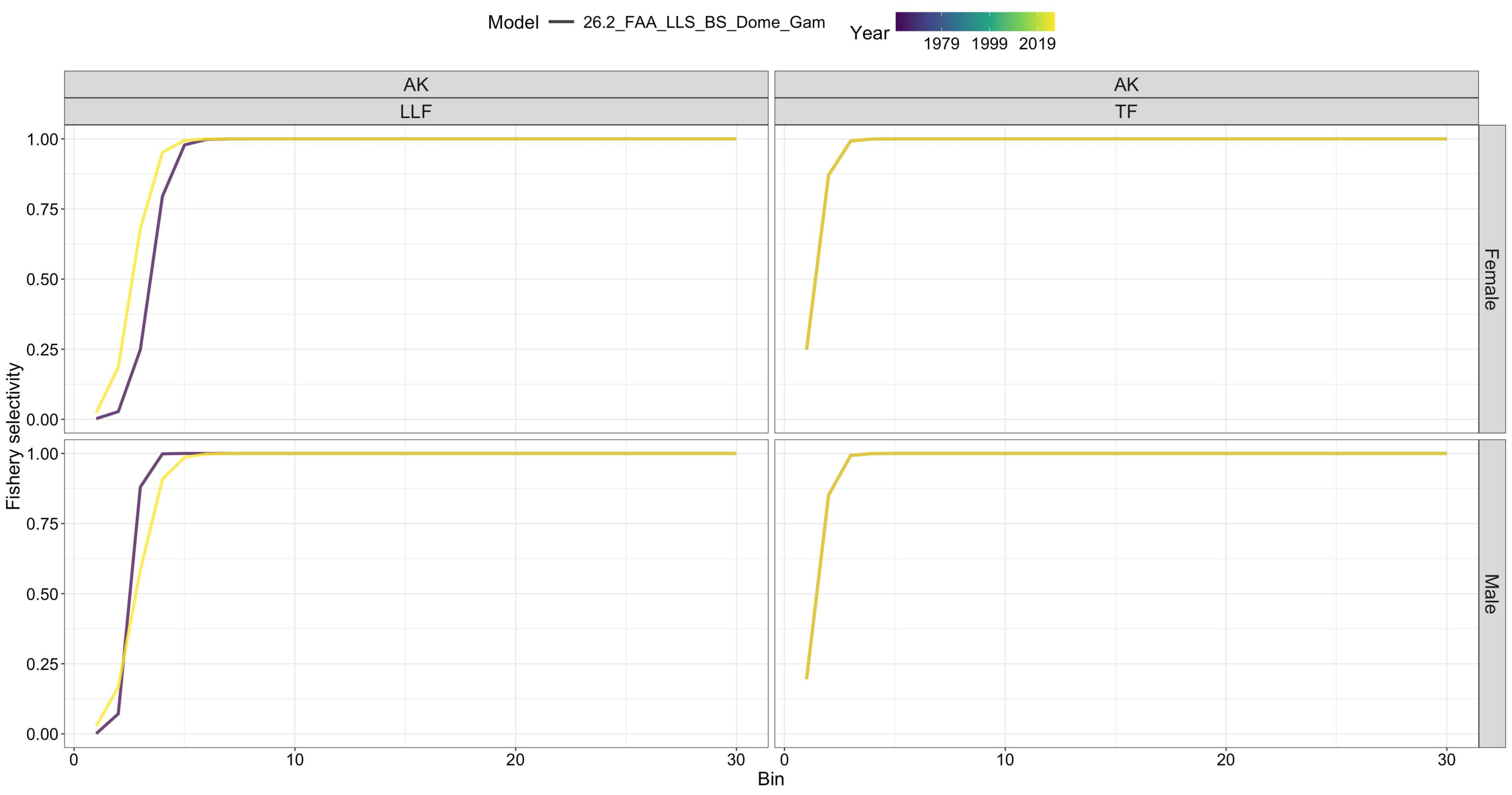
0
1
2
3
4
5
6
7

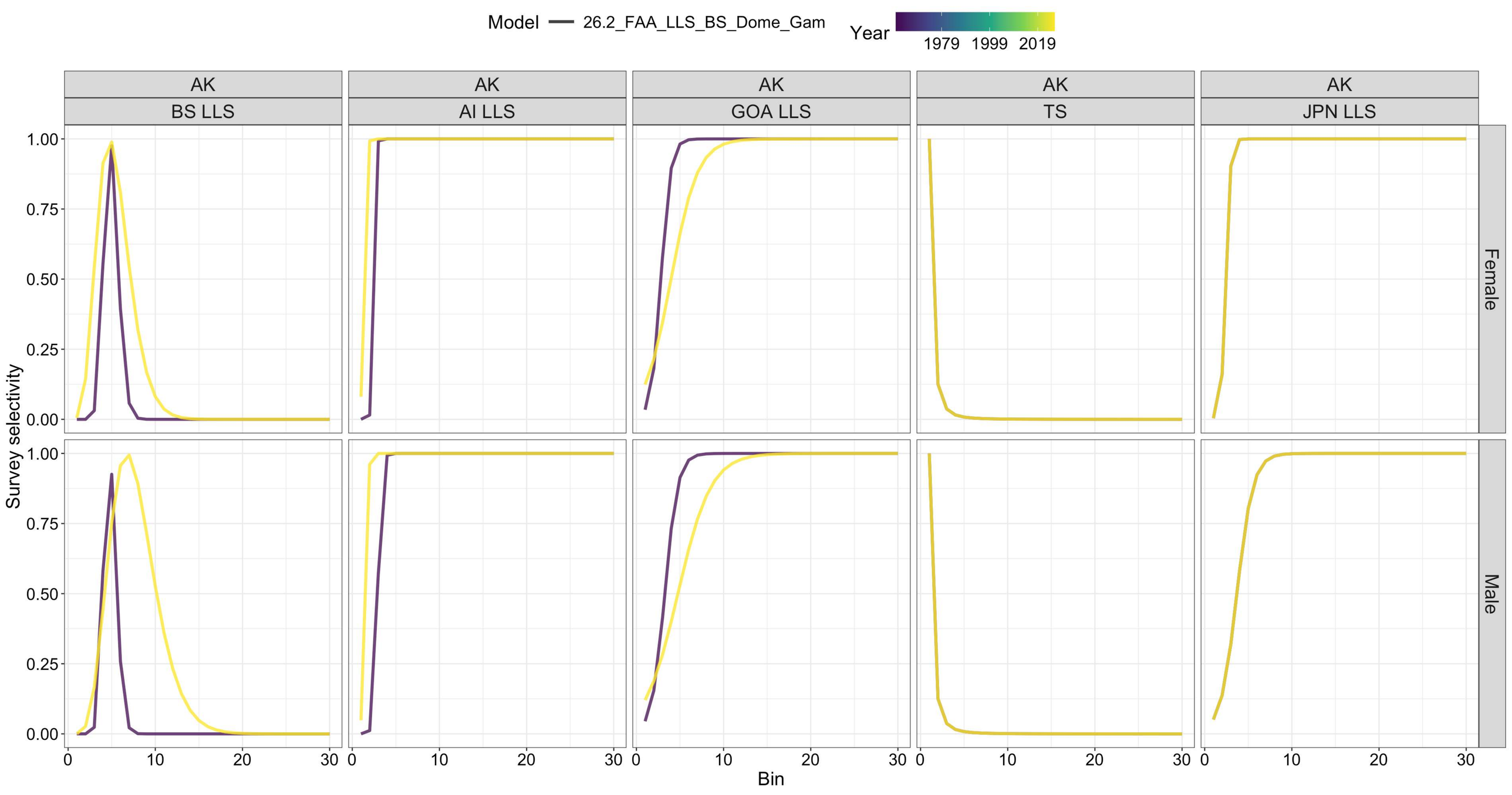
Years since cohort was last estimated

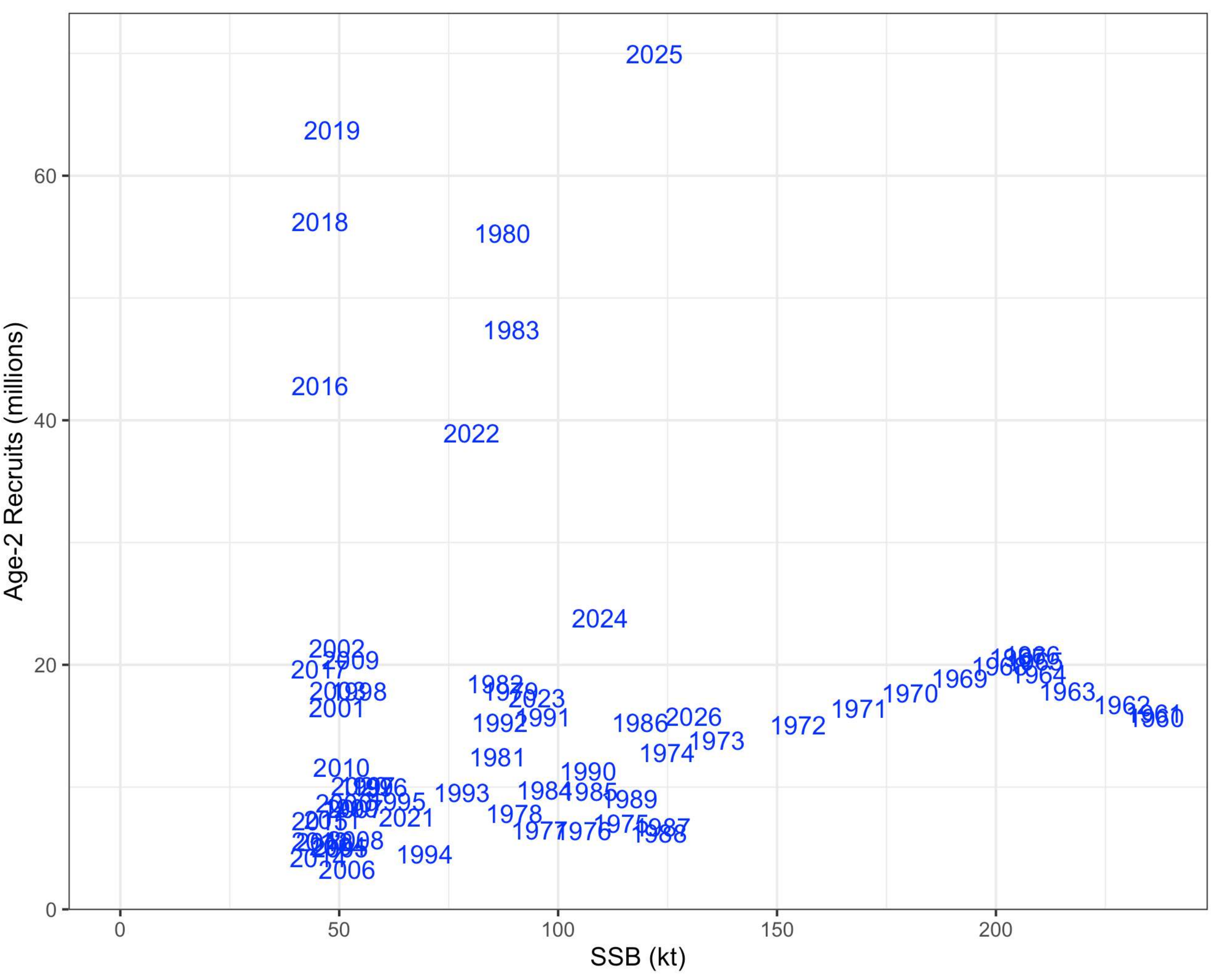
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Index Residuals Runs Test




Model
26.2_FAA_LLS_BS_Dome_Gam
Year
1979
1999
2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.2_FAA_LLS_BS_Dome_Gam
Year
1979
1999
2019
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Age-2 Recruits (millions)
SSB (kt)
0
20
40
60
0
50
100
150
200
2025
2019
2018
1980
1983
2016
2022
2024
2002
2017
2001
1982
1979
2023
1992
1991
1986
2026
1972
1971
1970
1969
1964
1963
1962
1973
1974
1981
2010
1990
1993
1984
1985
1989
1995
1978
2021
1977
1976
1975
1988
1994
2006

Population and Catch Trends

Recruitment (millions of fish)

SSB or Catch (kt)

Year

Catch

Recruitment

SSB

Model 26.2_FAA_LLS_BS_Dome_Gam

AK

Dynamic SSB0

Recruitment

SSB

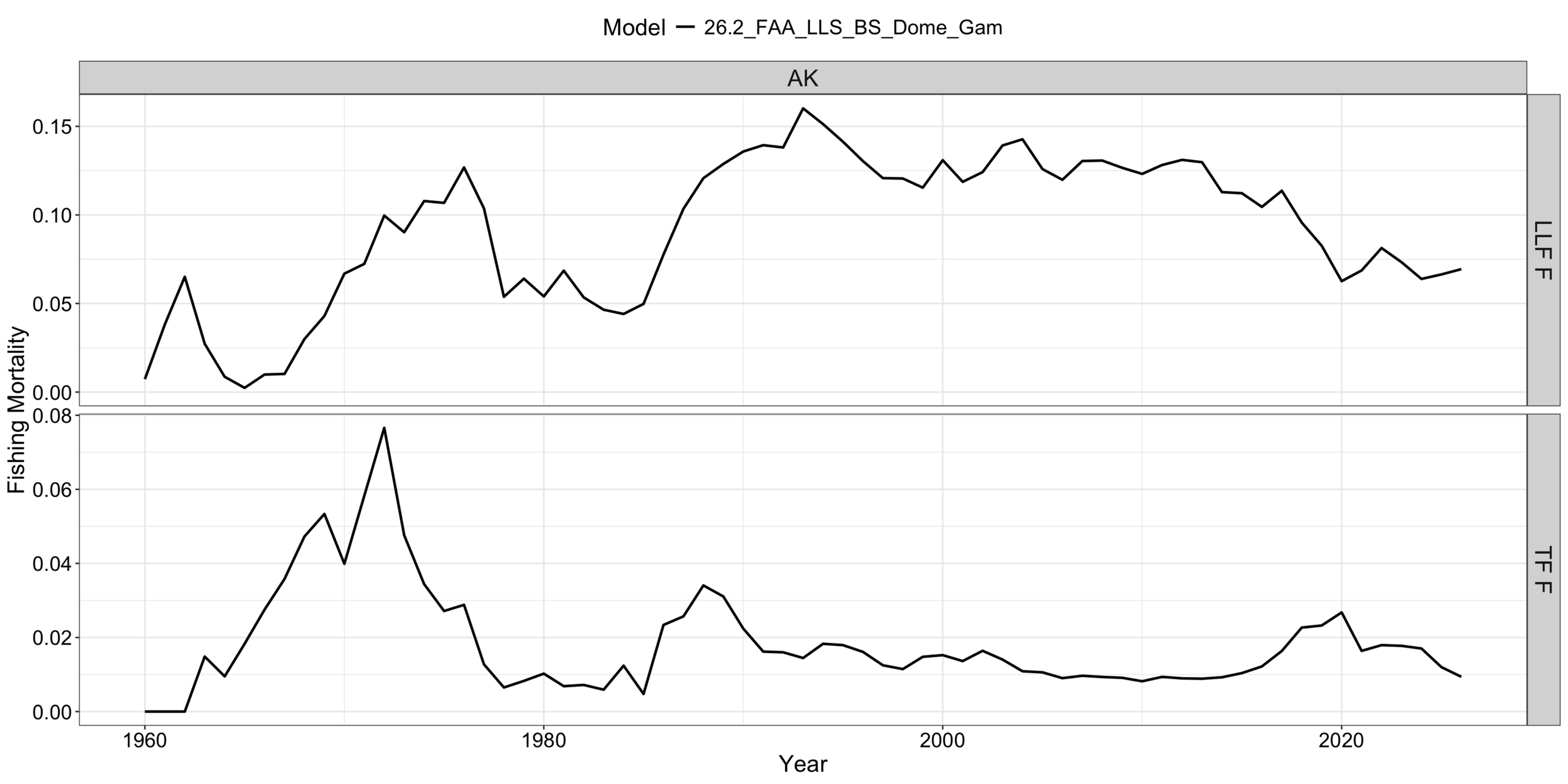
Total Biom

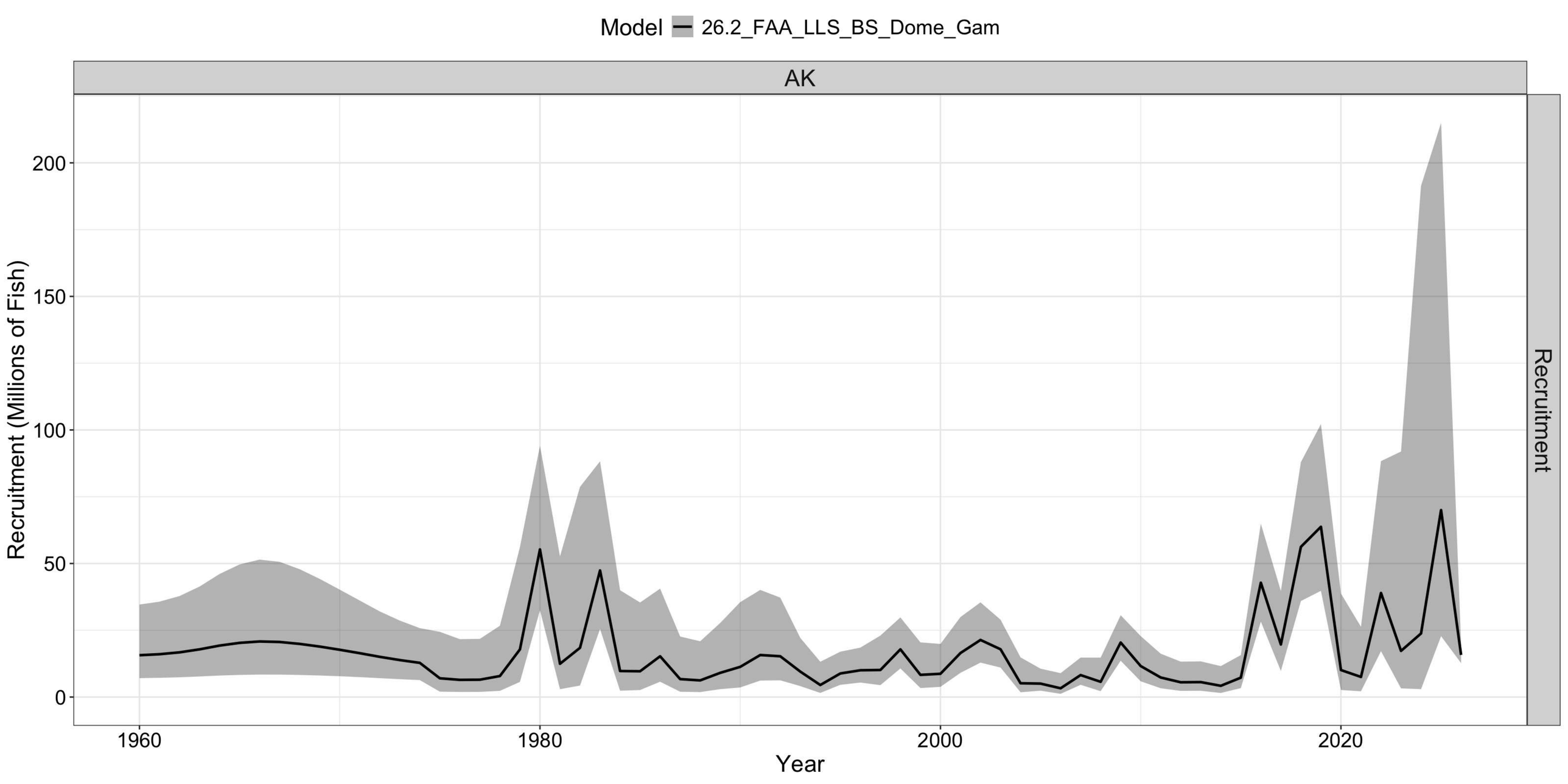
LLF F

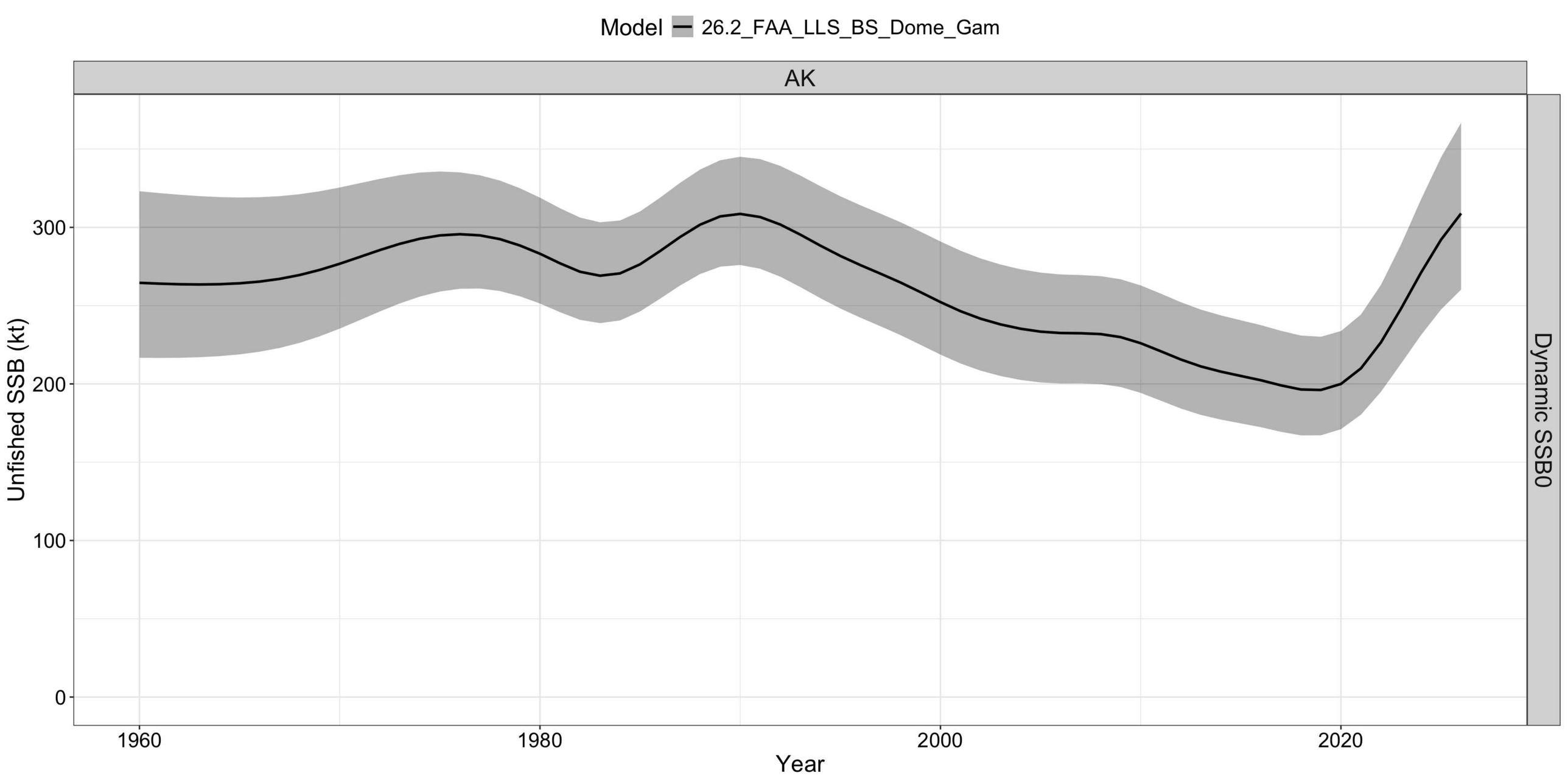
TF F

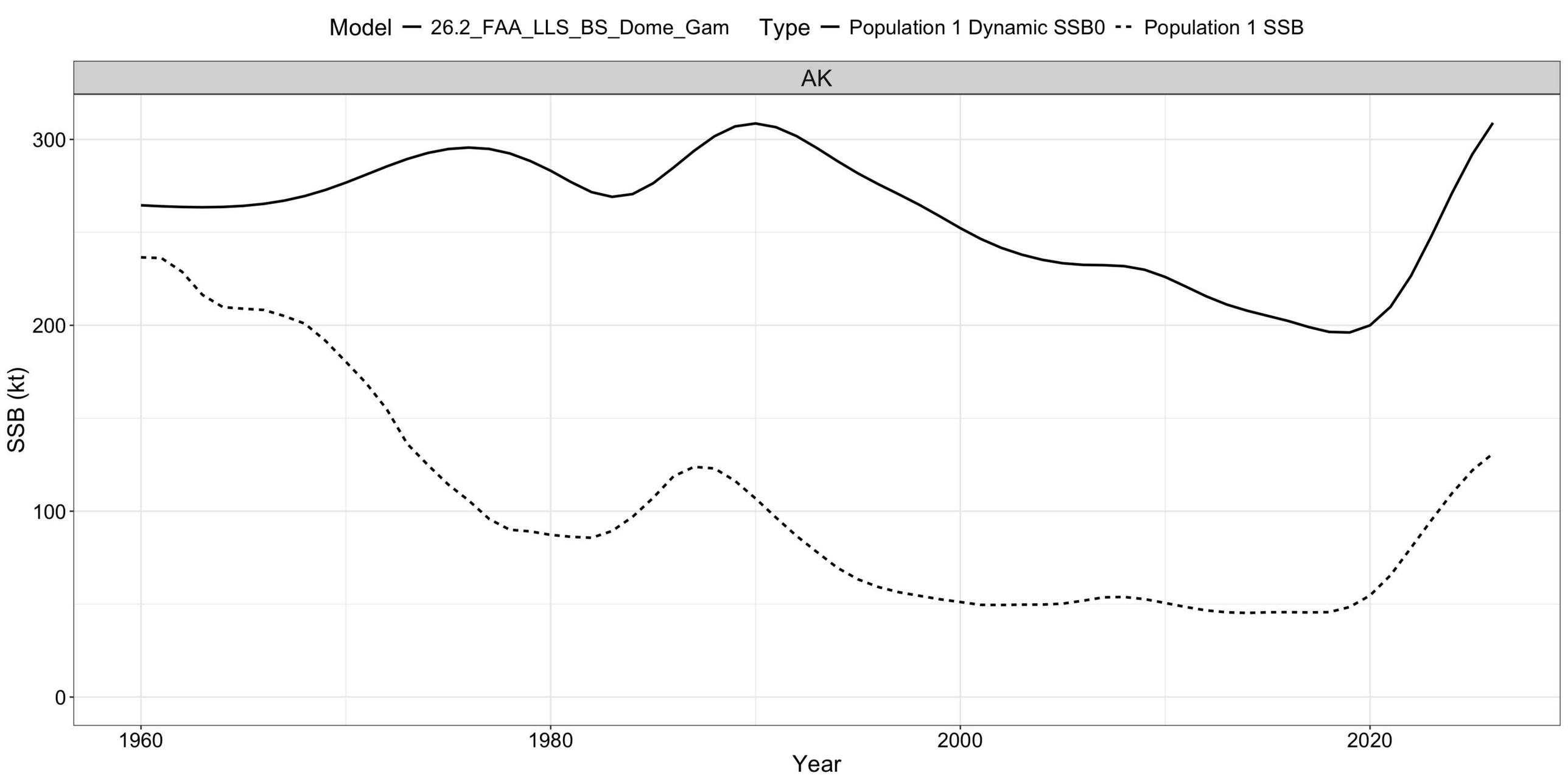
Value

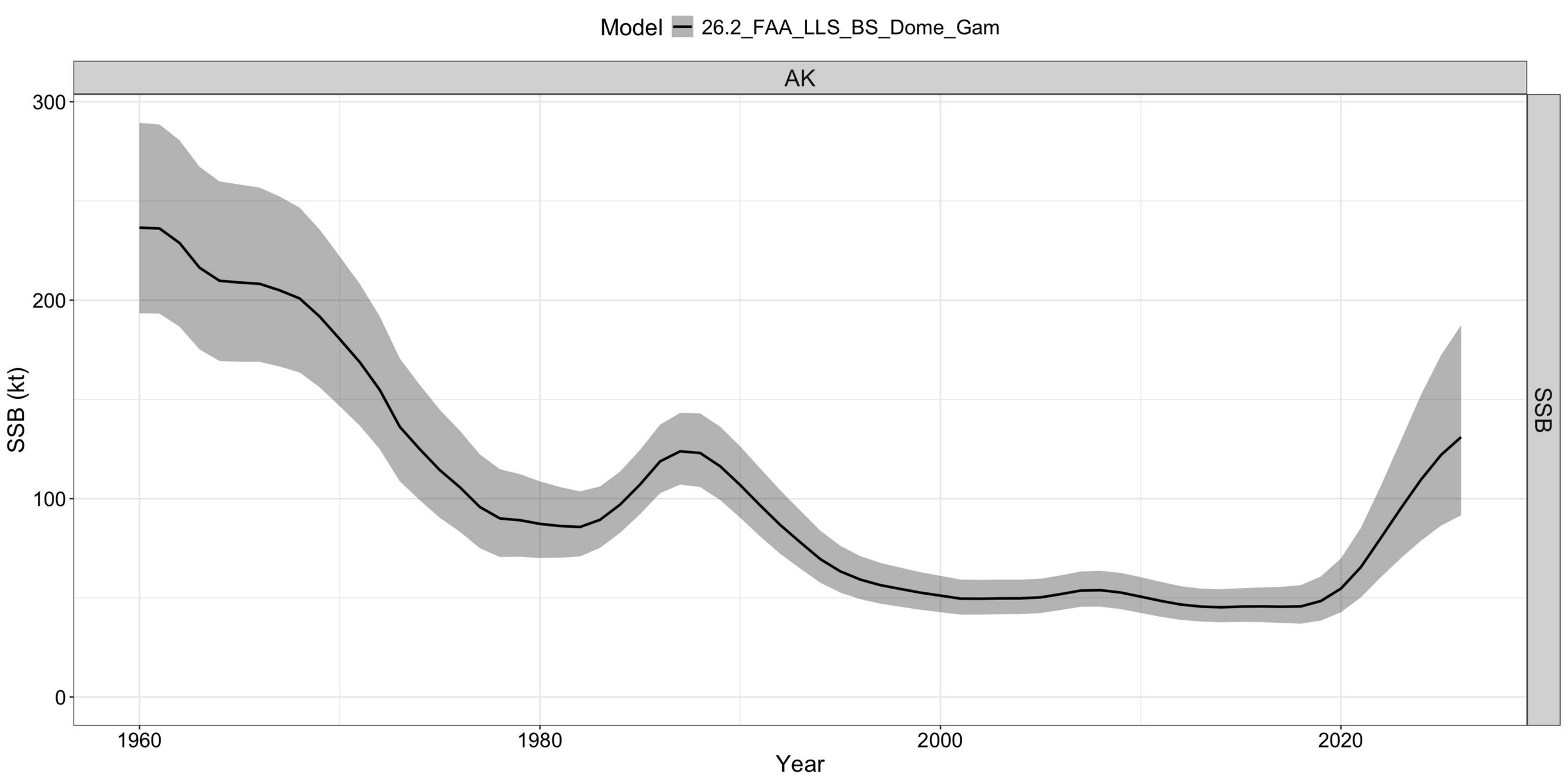
Year

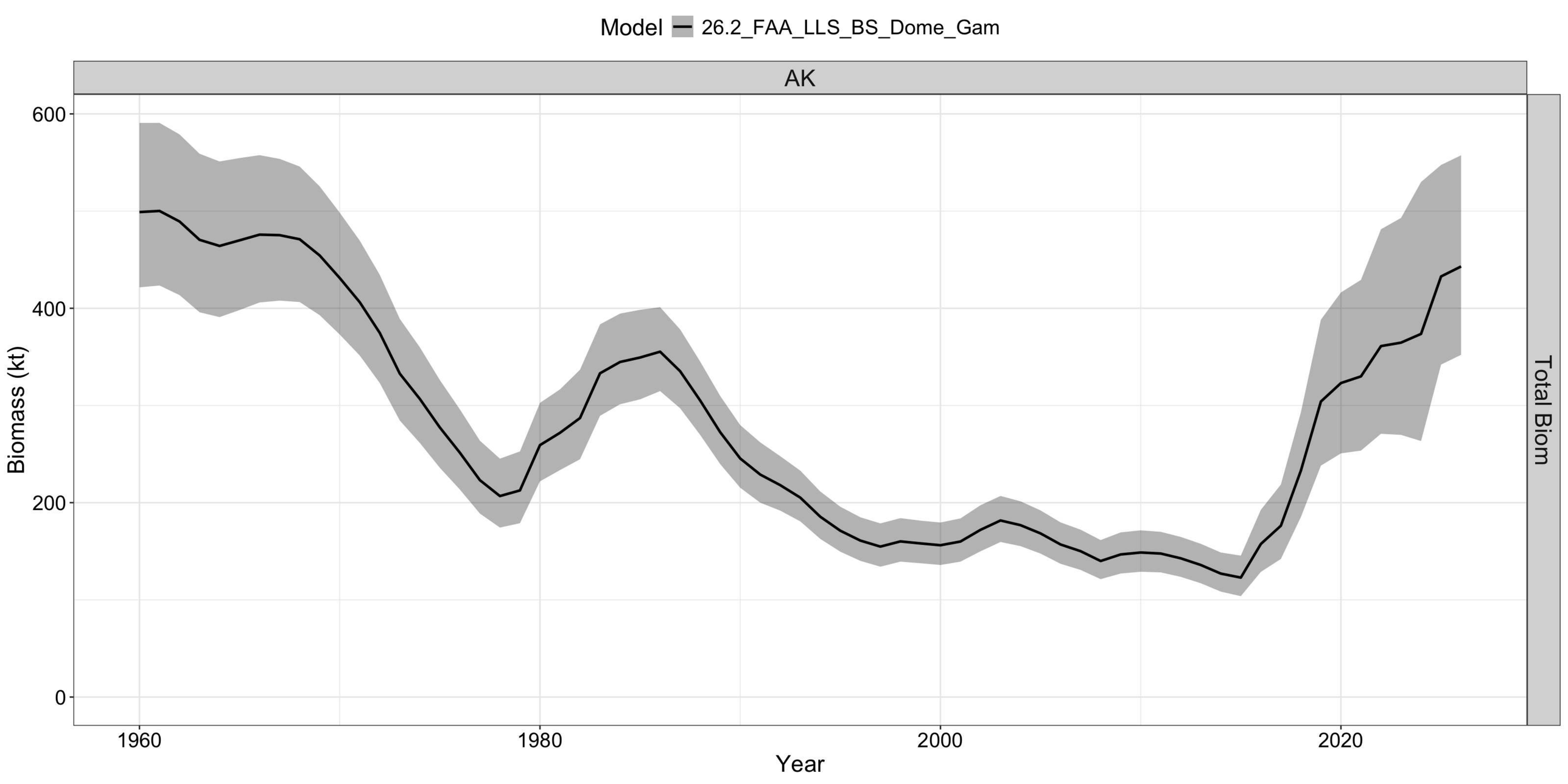

Model — 26.2_FAA_LLS_BS_Dome_Gam
AK
LLF F
TF F
Fishing Mortality
0.15
0.10
0.05
0.00
0.08
0.06
0.04
0.02
0.00
1960
1980
2000
2020
Year


Model
26.2_FAA_LLS_BS_Dome_Gam
AK
Recruitment (Millions of Fish)
200
150
100
50
0
1960
1980
2000
2020
Year
Recruitment

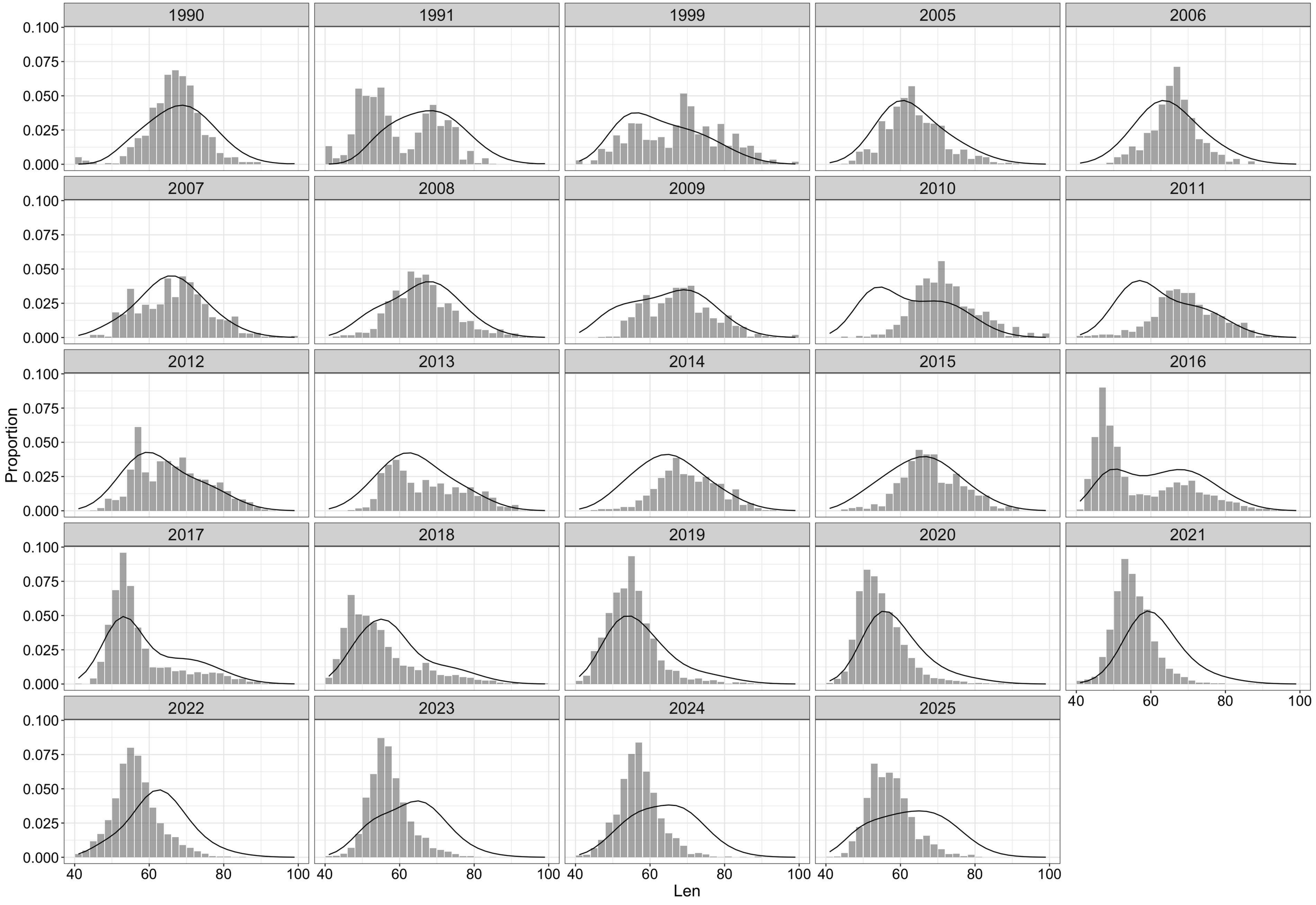

Model
26.2_FAA_LLS_BS_Dome_Gam
AK
Dynamic SSB0
Unfished SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year


Model — 26.2_FAA_LLS_BS_Dome_Gam
Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year

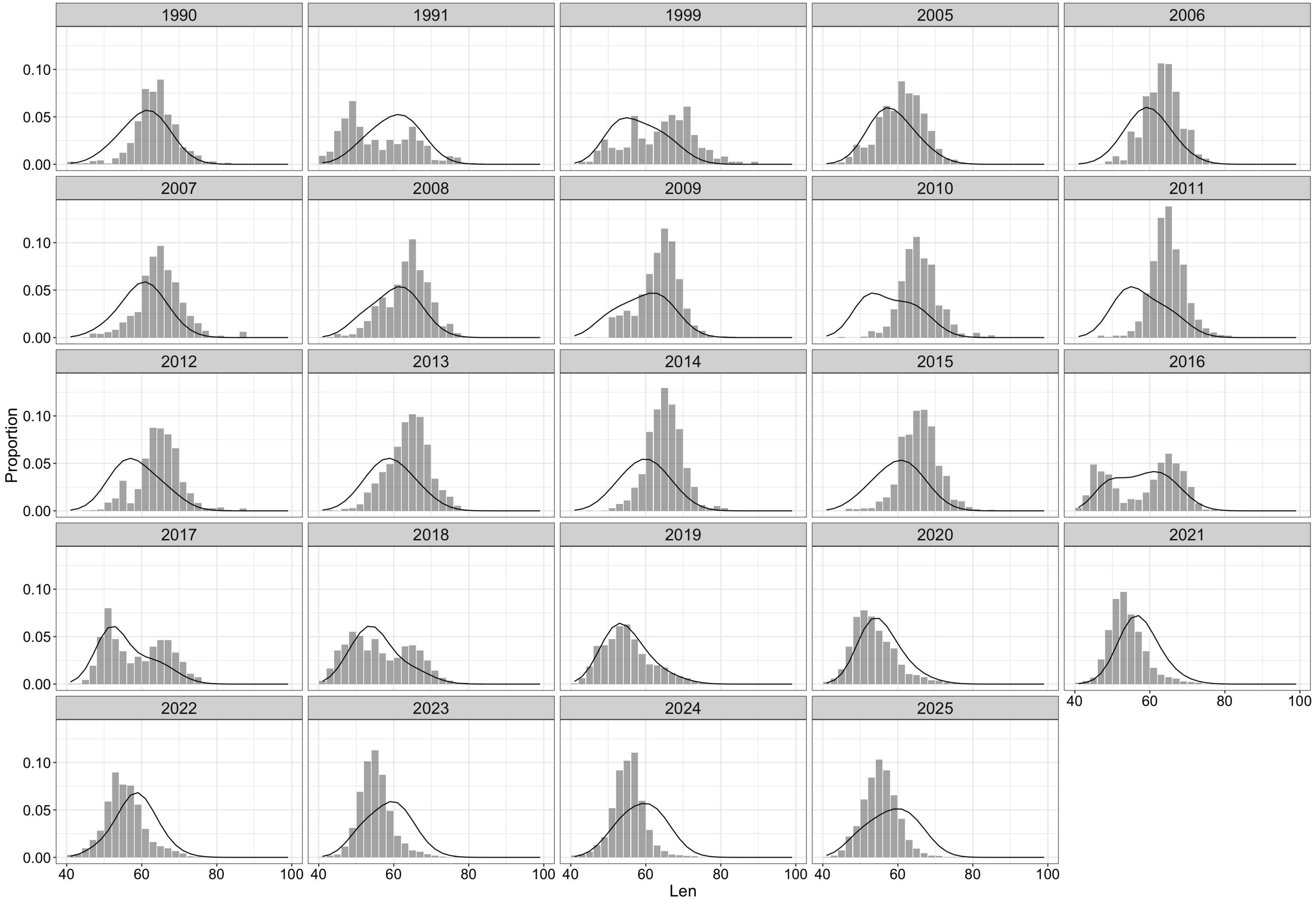

Model
26.2_FAA_LLS_BS_Dome_Gam
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year


Model
26.2_FAA_LLS_BS_Dome_Gam
AK
Total Biom
Biomass (kt)
600
400
200
0
1960
1980
2000
2020
Year

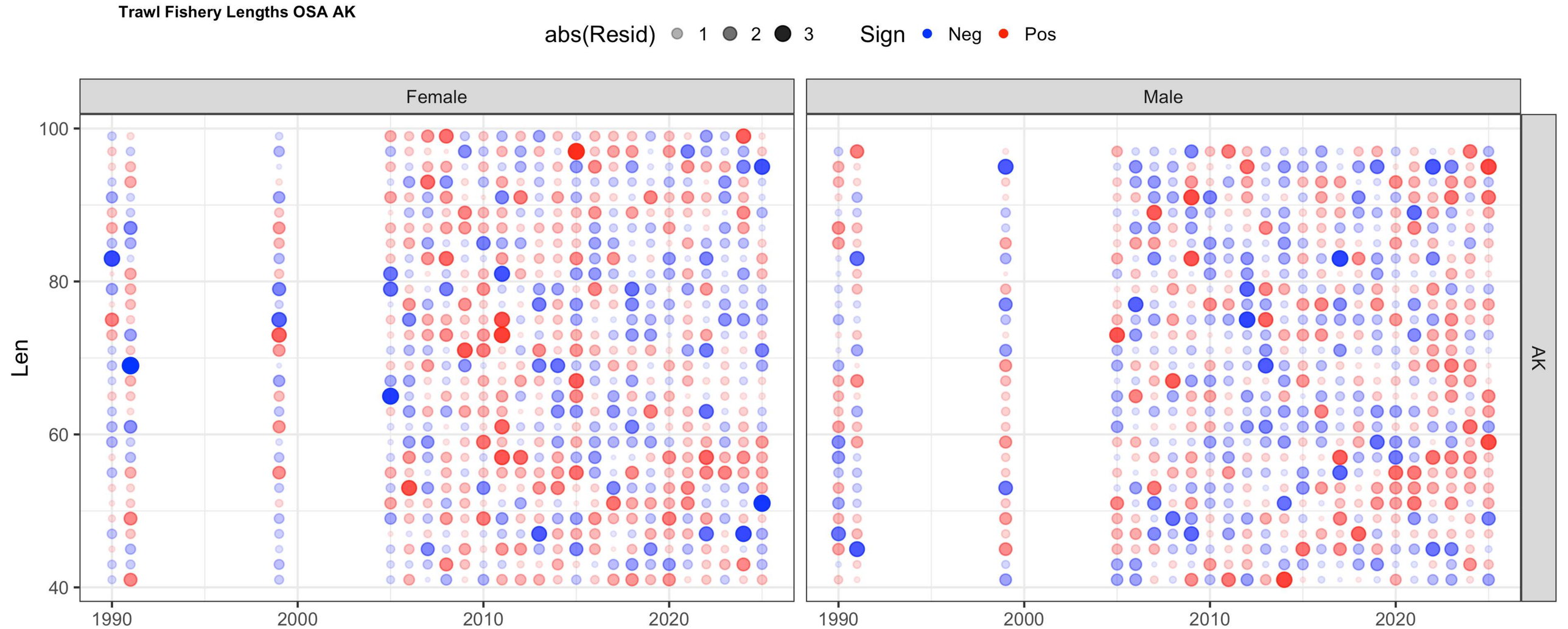
Trawl Fishery Lengths Female AK

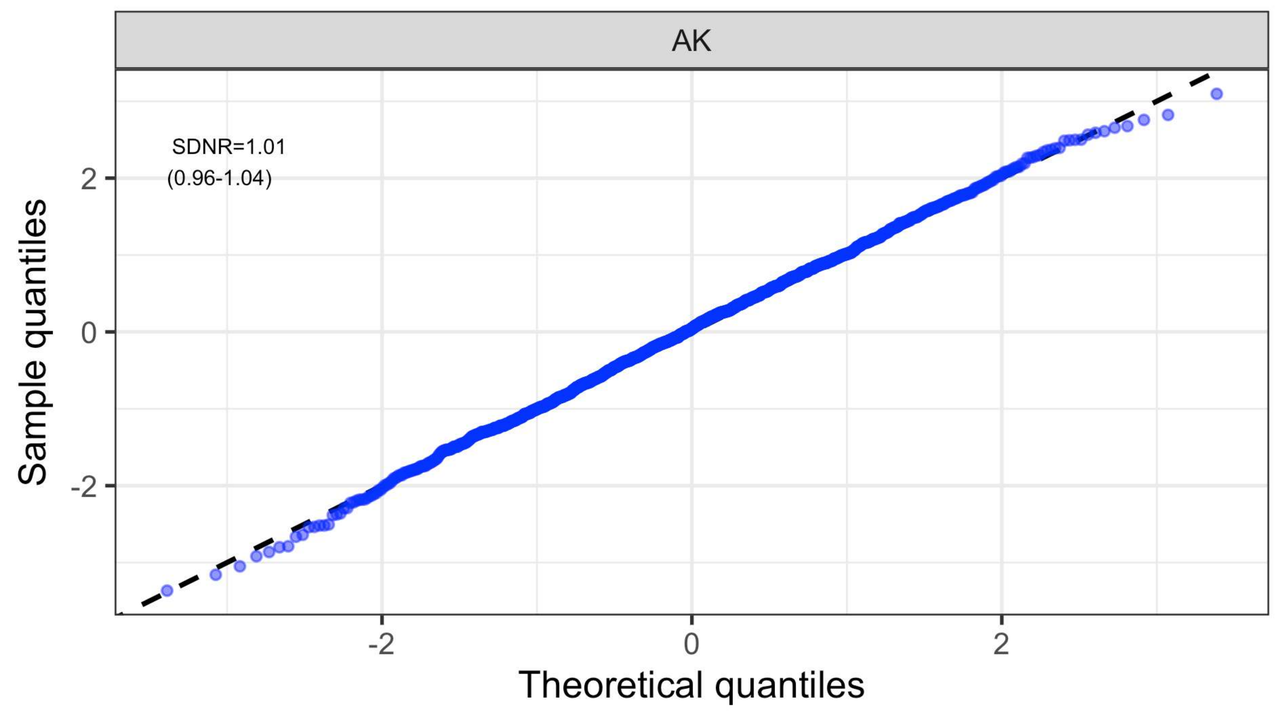


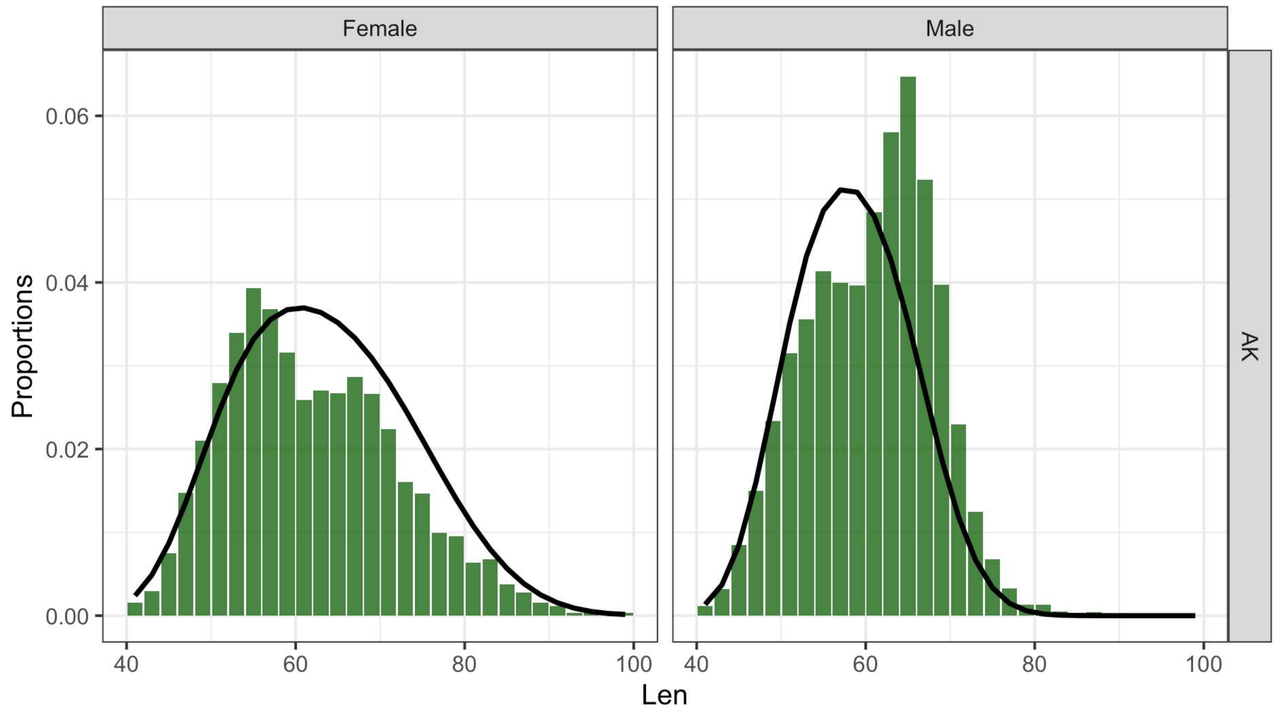
Trawl Fishery Lengths Male AK

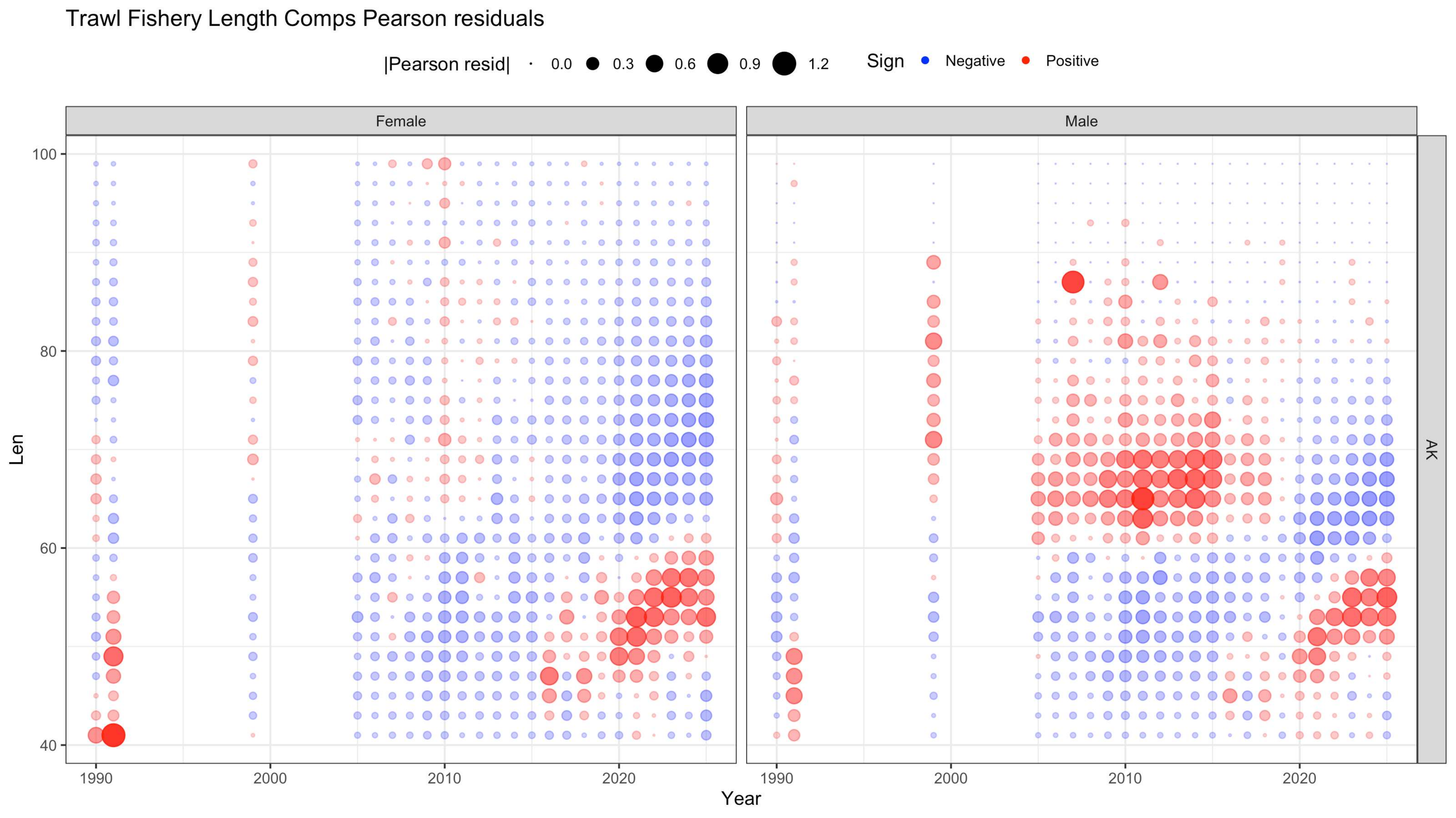


Trawl Fishery Lengths OSA AK

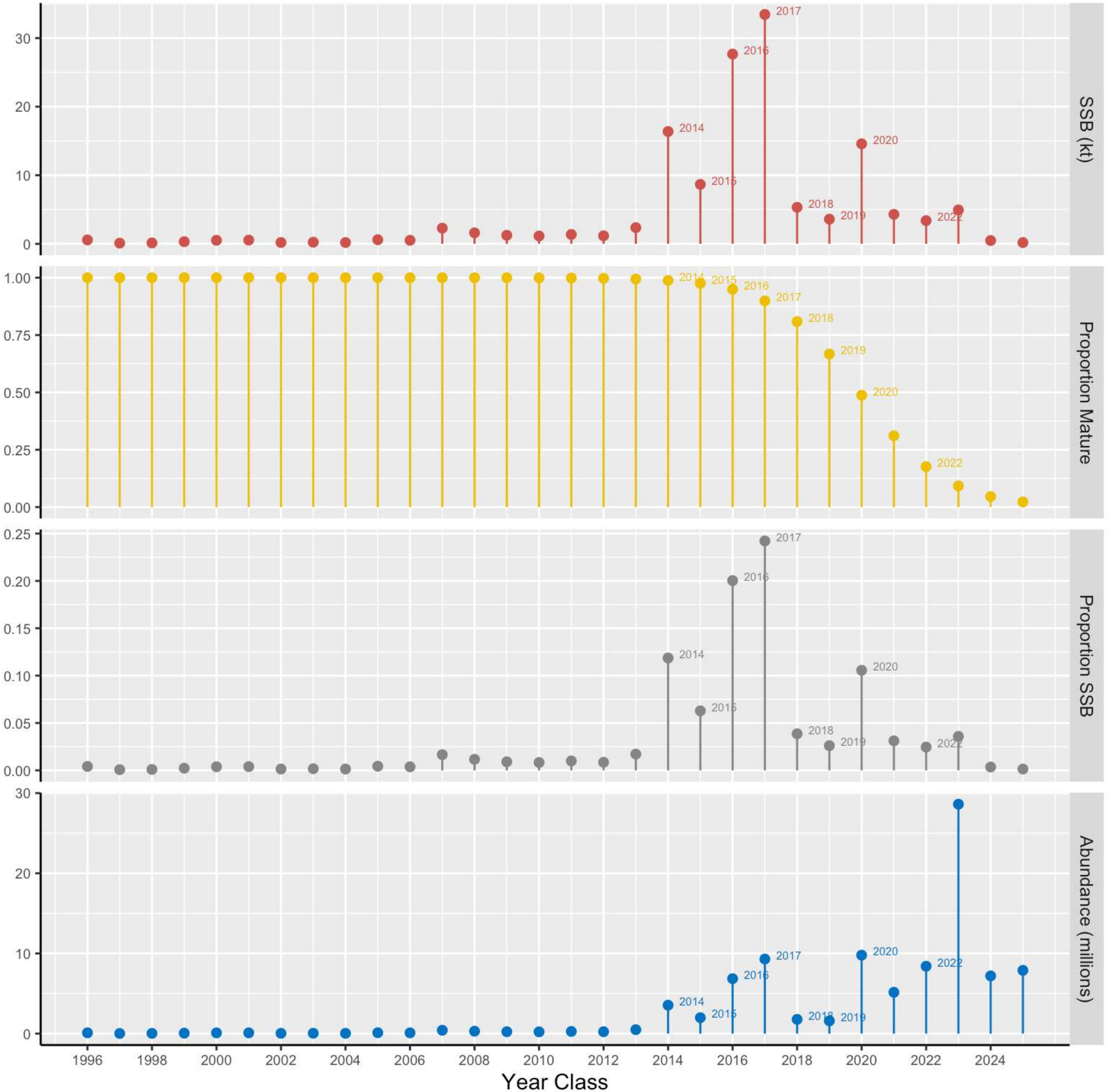






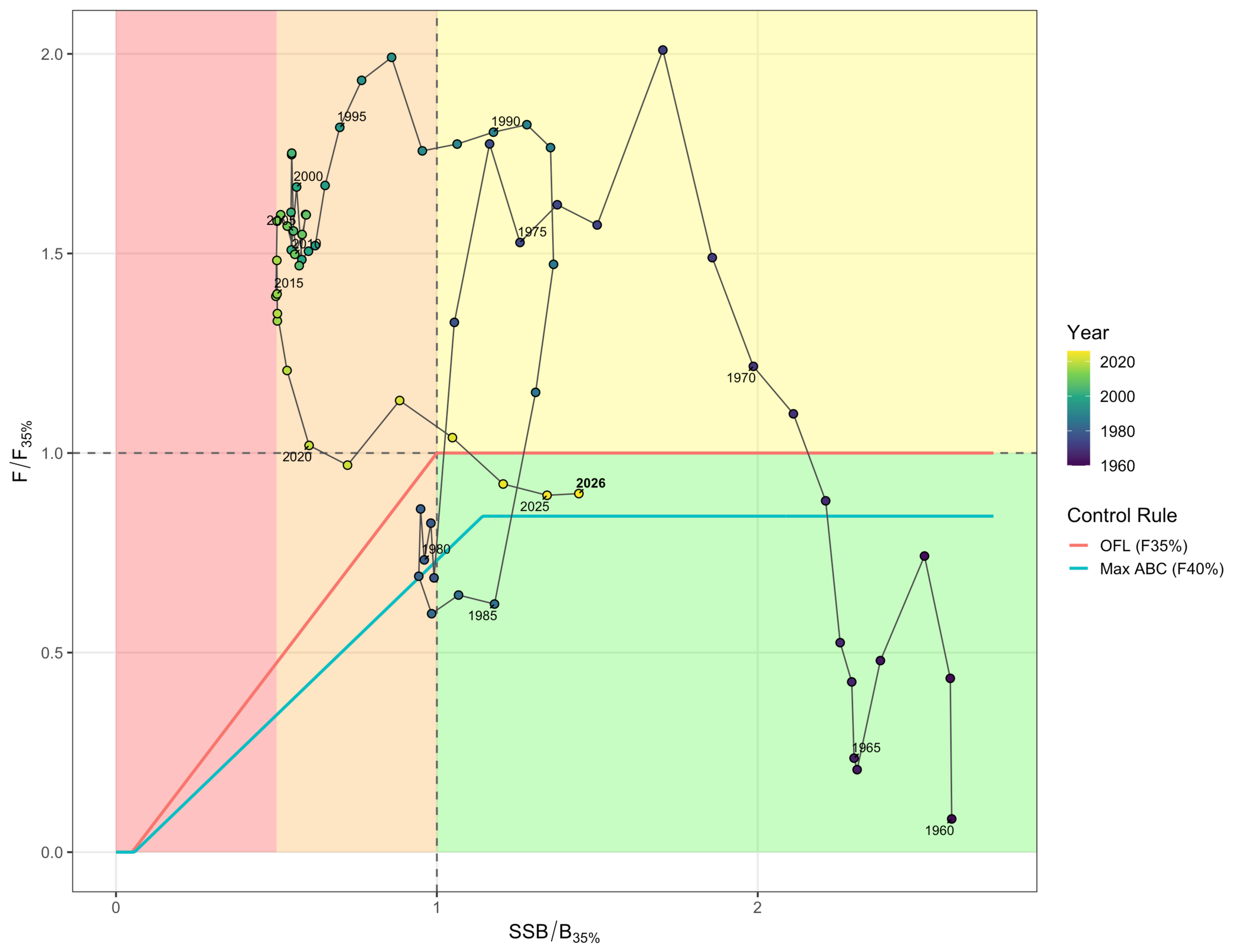

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.0
0.3
0.6
0.9
1.2
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

Contribution to 2027 SSB by Year Class



Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1.00	NA	NA
AI LL Survey Index	0.23	1.00	NA	NA
GOA LL Survey Index	0.08	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	NA	NA	NA	NA
BS LL Survey Ages	NA	0.01	34.07	0.33
AI LL Survey Ages	NA	2.15	35.57	76.52
GOA LL Survey Ages	NA	0.28	67.17	19.14
JPN LL Survey Ages	NA	0.18	83.33	14.69
JPN LL Survey Lengths	NA	0.33	54.62	18.06
Fixed Gear Fishery Ages	NA	0.05	142.85	6.47
Trawl Fishery Lengths	NA	0.10	47.62	4.65

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over
AA_LLS_BS_Dome_Gam	1	131.0215	308.9087	0.07878868	26.02088	114.8758	0.07383	1.14055	0.42414	1.06


2.0
1.5
1.0
0.5
0.0
0
1
2
F/F35%
SSB/B35%
1995
1990
2000
2005
2010
1975
2015
1970
2020
2026
2025
1980
1985
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.088	0.088
M Male (estimated)	0.114	0.114	0.09	0.09
Average Recruitment (millions)	25.9	25.9	16.4	16.4
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	447040	458366
Projected Female Spawning Biomass (t)	219,714	241,217	138107	150470
B100% (t)	302,672	302,672	287189	287189
B40% (t)	121,069	121,069	114876	114876
B35% (t)	105,935	105,935	100516	100516
FOFL	0.102	0.102	0.088	0.088
max FABC	0.087	0.087	0.074	0.074
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	30716	33201
OFLw (t)	58,532	57,797	30585	33070
max ABC (t)	50,283	49,651	26020	28131
ABC (t)	50,283	49,651	26020	28131
ABCw (t)	50,111	49,482	25889	28000
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

0.15

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0

200

400

600

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0

200

400

600

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0

20

40

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30

1960
1980
2000
2020
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30

2010
2020
2030
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
20
40
60

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
20
40
60

2010
2020
2030
2040

Year

S1: FABC (F40)
S1a: Author's F (FABC Ratio) Year 1, FABC After
S2: Author's F (FABC Ratio) Years 1+2
S3: 5-year Average F
S4: Half max FABC (1/2 * F40)
S5: No Fishing
S6: Overfished (FOFL)
S6a: Author's F (FABC Ratio) Year 1, FOFL After
S7: Approaching Overfishing (FABC -> FOFL)
SSB (kt)
0
100
200
300
1960
1980
2000
2020
2040
Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

2010

2020

2030

2040

Year